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How Does Pore Structure Affect the NMR Relaxation in Unsaturated Porous Media: A Simulation Study

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RESEARCH ARTICLE
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How Does Pore Structure Affect the NMR Relaxation in Unsaturated Porous Media: A Simulation Study

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Key Points:

- We develop a simulation framework to correct misinterpretation of unsaturated short nuclear magnetic resonance (NMR) T_2 where micropores exceed their saturation capacity
- Our simulation reveals that different pore shapes and pore size distributions induce distinct unsaturated NMR relaxation behaviors
- A new method to interpret NMR experimental data and evaluate subtle water variations in multi-pore systems at varied saturations

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Abstract Monitoring groundwater dynamics within the vadose zone is important for the investigation of many hydrological and ecological processes. Nuclear magnetic resonance (NMR) technology can be utilized to reveal these dynamics due to its unique sensitivity to water. The correlation between water content and distribution with NMR signals (relaxation times and amplitude) aids in discerning water retention patterns in porous media. However, interpreting NMR data to understand unsaturated pore-water dynamics is challenging due to complex pore environments and fluid-rock-air interactions. Especially, previous studies often misinterpret the increased amplitude of shorter relaxation T_2 components of unsaturated T_2 to erroneously imply that the small pore exceeds their maximum saturation capacity. We develop a simulation framework to accurately track pore-water dynamics and NMR responses (T_2 and T_1-T_2) in unsaturated multi-pore systems. Dual-spherical and dual-triangular pore systems with different pore size distributions modeled the imbibition process. Simulations clarified the decrease in shorter relaxation T_2 components in unsaturated states, revealing that unsaturated macropores and saturated micropores can exhibit the same short relaxation times when their fluid volume to fluid occupied surface area ratio is identical. Our simulation also demonstrates that different pore shapes and pore size distributions lead to distinct unsaturated NMR signals. Furthermore, we propose a new method evaluating water distribution in various natural porous media at various saturations by interpreting experimental unsaturated NMR data. Our work enhances unsaturated NMR data interpretation, providing accurate insights into water distribution and pore structure in unsaturated porous media.

Plain Language Summary Groundwater is a vital resource retained in soil and weathered rock beneath the surface. Monitoring this subsurface water storage and movement is crucial for managing water resources for drinking, agriculture, and sustaining ecosystems. Nuclear magnetic resonance (NMR) is a powerful geophysical tool well-suited for studying water in soil and rock due to its unique sensitivity to water molecules. While NMR can provide detailed insights into water content and distribution within saturated soils and rocks, interpreting NMR signals from unsaturated subsurface materials remains challenging, even at the laboratory scale. To address this challenge, we developed a simulation framework to model NMR responses from unsaturated pores. First, our modeling results correct a misinterpretation of shorter NMR relaxation time in unsaturated conditions. The results also show that the different pore sizes and shapes can cause very distinct unsaturated NMR relaxation times. This study also proposes a new NMR analysis method to evaluate water distribution in unsaturated soil and rock, correcting the misinterpretation of unsaturated experimental NMR data in previous studies. Our work enhances the understanding of the NMR applications in unsaturated conditions, and our approach overcomes limitations in previous studies that may have mischaracterized water distributions in partially saturated subsurface.

1. Introduction

Groundwater plays an important role in the global water cycle, climate change (e.g., Amanambu et al., 2020; Brunner et al., 2017), and other numerous human and engineered processes. Beyond fully saturated aquifers, groundwater also includes the vadose zone, where water is retained in unsaturated soils as soil moisture or trapped as rock moisture in weathered bedrocks. The vadose zone is vital to various hydrological processes, such as water seepage, water evaporation from soil and rock, seasonal freeze-thaw cycle of soil, etc. Hydrodynamics are integral in many ecological and human activities, ranging from the water intake of plants from the soil via roots to pumping for irrigation and drinking water. Consequently, alterations in the vadose zone's water content and water dynamics could trigger issues such as swampiness, salinization, and desertification, facilitate the transport of contaminants, or potentially lead to geological hazards like landslides and land subsidence (Alsubal et al., 2018; Ren et al., 2021; Tang et al., 2021). The pore water distribution within natural porous media can affect the

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mechanical properties, damage characteristics, and other properties of rock and soil (e.g., Zhou et al., 2022) and oil and gas shale reservoirs evaluation (e.g., Li et al., 2020; Li et al., 2021). Therefore, understanding groundwater dynamics within the vadose zone is essential for the effective management of groundwater resources, ecosystem preservation, and disaster mitigation.

Various measurement methods and techniques have been employed to understand the water retention and hydrological processes in the vadose zone. Traditional vadose zone studies often involve extracting soil samples for laboratory analysis of hydraulic properties through some methods such as time-domain reflectometry (TDR), neutron moisture gauges, and tensiometers, which can also be applied in situ. For instance, tensiometers, consisting of a porous ceramic cup in soil, directly measure matric potential and can determine water properties like retentivity, conductivity, and diffusivity across different saturation levels using various setups (e.g., Dane & Topp, 2002; Dirksen & Matula, 1994; Kutílek & Nielsen, 1994). Advanced soil moisture measurement techniques, such as the infrared moisture balance method, supported by the microprocessor, integrate electrobalancing with infrared heating to rapidly assess soil moisture content in small sample sizes (Robinson et al., 2008; Terhoeven-Urselmans et al., 2008). Nonetheless, scaling these methodologies for field applications presents challenges due to their limited sensitivity with small samples, and logistical and financial obstacles in deploying sensor networks over large areas and depths. Additionally, the intrusive nature of sampling potentially alters the soil's natural state, leading to a loss of in situ hydrological data.

Geophysical methods characterize the spatiotemporal distribution of subsurface properties in minimally invasive ways. In vadose zone research, the application of geophysical methods such as ground penetrating radar (GPR) and electrical resistivity is well established (e.g., Binley et al., 2015). The electromagnetic and electrochemical signals are sensitive in capturing variations in water content and flow across spatial scales, providing imaging for dynamic and static hydrological processes and elucidating the hydrodynamics of the vadose zone's soil and rock formations. Previous studies used such geophysical tools to detect soil moisture variations beneath trees (e.g., Huth & Poulton, 2007), to map soil-water holding capacity (e.g., Abdu et al., 2008), and to differentiate moisture content across soil types (e.g., Martinez et al., 2014).

Among these methods, low-field nuclear magnetic resonance (NMR) stands out due to its unique sensitivity to water (Coates et al., 1999; Khetrapal et al., 1975; Prammer, 2004), making it a powerful tool in hydrological studies across diverse environments. Low-field NMR utilizes radiofrequency pulses applied to protons within a magnetic field and measures the response of their magnetic moments. It is based on the interaction of magnetic moments of protons in an applied magnetic field and the relaxation behaviors of atomic nuclei after removing the applied field. By leveraging the magnetization and relaxation of hydrogen atoms' spin magnetic moments, the magnitude and distribution of recorded NMR relaxation time (T_1 or T_2) provide estimates of water content, pore size distribution, and hydraulic conductivity (Behroozmand et al., 2015). NMR tools, including high-resolution laboratory benchtop devices, along with field instruments deployed either at the surface (sNMR) or in the borehole (bNMR) to capture the relaxation responses from subsurface water are applied in hydrogeophysical applications (e.g., Guérin, 2005; Jiang et al., 2020; Kirkland & Codd, 2018), offering invaluable insights into water dynamics in soil and rocks underground (e.g., Behroozmand et al., 2015; Flinchum et al., 2019; Legchenko & Valla, 2002).

The majority of NMR hydrogeophysical studies have focused on the saturated zone. In recent years, there has been growth in both laboratory and field-based research employing NMR in the unsaturated zone. One example is the utilization of laboratory NMR to characterize the soil-water characteristic curve (SWCC) in unsaturated soils (e.g., Dong et al., 2023; Kong et al., 2018; Teng et al., 2020). Additionally, both surface and borehole NMR techniques have been employed to determine water content within the vadose zone in the field (e.g., Flinchum et al., 2019; Legchenko & Valla, 2002; Rempe et al., 2018). The relaxation behaviors of protons in unsaturated samples are influenced by saturation levels, pore structure, and physicochemical interactions at fluid-rock interfaces. By capturing these nuanced changes, NMR provides insights into unsaturated pore-water dynamics that were previously difficult to quantify. These are crucial for understanding dynamic groundwater recharge-discharge processes, the imbibition and drainage of soil and rock moisture, and the distribution and partitioning of moisture within soil and rock (Schmidt & Rempe, 2020; Tian & Wei, 2020; Yang et al., 2019; Zhang & Zhang, 2021).

However, the interpretation of NMR data within the vadose zone presents a complex challenge due to the zone's inherent heterogeneity. Established relationships between NMR signals, pore structure, and physicochemical

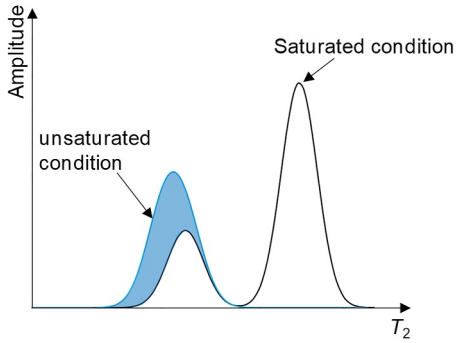


Figure 1. Comparative NMR T_2-distribution in unsaturated and saturated porous media (the shaded blue area signifies the reduced amplitude of short relaxation times from unsaturated to saturated conditions).

properties, while reliable in saturated conditions, often fail or result in significant misinterpretation and errors when applied to unsaturated conditions. This is primarily due to the complex architecture of vadose zone soil and rocks, as well as various solid-liquid-vapor interactions within pore space. In porous media with dual- or multi-mode pores, the water distribution during imbibition or drainage is difficult to quantify from NMR experimental data. Previous studies have shown that unsaturated conditions lead to a shift in the NMR T_2 distribution curve toward shorter relaxation times, outside the curve envelope of saturated conditions (e.g., Bird et al., 2005; Meyer et al., 2018), as illustrated in Figure 1. Such shifts have been mistakenly attributed to increased water content in micropores or bound (i.e., absorbed on the pore wall) water content within unsaturated media compared to their saturated status (e.g., Cocusse et al., 2022; Peng et al., 2023; Zhang & Zhang, 2021; Zou et al., 2024). This interpretation is fundamentally flawed and counter-intuitive, given that smaller pores cannot contain more water than their maximum saturation capacity. Such data misinterpretation suggests a fundamental misunderstanding of unsaturated water distribution from NMR data, underlining the need for a new approach to the analysis of NMR data.

Consequently, interpreting NMR data to evaluate water distribution, pore size, and pore shape in unsaturated states becomes a non-trivial and challenging task. To better interpret the unsaturated NMR responses, many previous studies have also utilized pore models with various shapes, such as the triangular pore (e.g., Mohnke et al., 2015), cylindrical pore (Falzone & Keating, 2016a), and interparticle pore throat of packed spheres (e.g., Mitchell et al., 2019), to present the wetting or drying of porous media and clarify its unsaturated NMR relaxation behaviors. However, no study has shown the effect of different pore sizes and shapes on unsaturated NMR relaxation behaviors, contributing to interpretation challenges of unsaturated NMR responses. Addressing these interpretation challenges due to complex pore structures is crucial to fully leverage NMR's potential in vadose zone research.

In this study, we aim to resolve the unsaturated NMR signal interpretation challenges through the following objectives: (a) we develop a simulation framework to simulate pore-water dynamics and their NMR responses within multi-pore systems; (b) we implement numerical simulations to correct the aforementioned misinterpretation in unsaturated NMR experiment data with multi-pore; (c) we investigate the evolution of NMR relaxation behaviors and evaluate water distribution in multi-pore systems with various pore shape, pore size, and pore distribution at various saturations. Our results resolve critical interpretation issues of unsaturated NMR data, which can substantially improve the interpretation of field NMR measurements and advance our understanding of vadose zone groundwater dynamics.

2. NMR Theory and Measurements

The NMR technique is a cost-effective, non-invasive, and novel technology for investigating natural porous media (e.g., soil and rock), and is often used for pore structure characterization, fluid identification, and petrophysical property testing. Low-field ^{1}H NMR measures the nuclear spin signals of hydrogen protons residing in porous media. When the NMR experiment starts, a constant external magnetic field (B_0) is applied to align magnetic spins of hydrogen nuclei to an equilibrium state of precession at Larmor frequency. Then a second radio frequency magnetic field (B_1) perturbs this alignment. Upon removal of this secondary magnetic field, the excited nuclear spins relax back to their equilibrium state.

The relaxation process can be expressed as a multi-exponential decay in the transverse direction (T_2) with respect to the static magnetic field (i.e., in the xy-plane) and a multi-exponential growth in the longitudinal direction (T_1) aligning with the static magnetic field (i.e., in the z-direction) (Brownstein & Tarr, 1979)

$$M_{xy}(t) = M_0 \sum_i f_{2i} e^{-\frac{t}{T_{2i}}} \tag{1}$$

$$M_z(t) = M_0 \left(1 - \sum_i f_{1i} e^{-\frac{t}{T_{1i}}}\right) \tag{2}$$

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here, $M_{xy}(t)$ and $M_z(t)$ are the transverse and longitudinal components of the NMR signal; M_0 is the initial signal magnitude and is proportional to the number of protons in the measured sample. f_{2i} is the proportion of the magnetic field relaxing in the transverse direction with relaxation time T_{2i}, and similarly, f_{1i} is the proportion of the magnetic field relaxing in the longitudinal direction with relaxation time T_{1i}. The values of T_{2i} are often plotted versus f_{2i} to yield the T_2-distribution.

The initial magnitude of the relaxation decay is proportional to the hydrogen content, whereas the T_1 or T_2 relaxation is dominated by three mechanisms: surface (T_{1S}, T_{2S}), bulk (T_{1B}, T_{2B}), and diffusion relaxation (T_{2D}). Surface relaxation is induced by protons diffusing inside pore space and encountering the pore surface. Bulk relaxation takes place spontaneously in the fluid and depends on fluid properties. Diffusion relaxation occurs in inhomogeneous magnetic fields and only affects T_2. These relaxation mechanisms can be expressed as:

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1B,2B}} + \frac{1}{T_{1S,2S}} + \frac{1}{T_{2D}} \tag{3}$$

In geologic media such as soil and rock without internal inhomogeneous magnetic fields (without diffusion relaxation), bulk relaxation only depends on fluid properties, and the surface interactions at solid-fluid interfaces accelerate the rate of NMR relaxation in the confined system as porous media, implying that only the surface relaxation mechanism affects the T_1 or T_2 relaxation. For confined saturated pore space, parameter κ (Brownstein & Tarr, 1979) is introduced to quantify relaxation due to the interaction of the protons with the pore wall and can be written as:

$$\kappa = \frac{\text{relaxation rate}}{\text{diffusion rate}} = \frac{\rho_s/l}{D/l^2} = \rho_s \frac{l}{D} \tag{4}$$

where ρ_s is the surface relaxivity which reflects the molecular interactions at the solid-fluid boundary with the typical value between 1 and 50 μm/s (Mohnke & Klitzsch, 2010), D is the diffusion coefficient, and l is the average distance traveled by a proton to reach the paramagnetic site during the NMR measurement, which is governed by the pore size. κ can be used to define fast, intermediate, and slow diffusion regimes (further considerations regarding diffusion regimes can be found e.g., in Müller-Petke et al., 2015). Fast diffusion (also called surface-limited), with $\kappa << 1$ given by Brownstein and Tarr. (1979) ($\kappa < 0.1$ given by Ryu, 2009), represents that the diffusion rate is higher than the relaxation rate. It means that the relaxing spins from the pore surface, where relaxation occurs, can diffuse throughout the pore space within the timescale of NMR measurement (e.g., Falzone & Keating, 2016a; Washburn, 2014). For a system in the fast diffusion regime, bulk relaxation is slower than surface relaxation and can be often neglected. Therefore, Equation 3 can be written as:

$$\frac{1}{T_{1,2}} \approx \frac{1}{T_{1,2S}} = \rho_s S_{por} \tag{5}$$

where S_{por} is the pore's surface area-to-volume ratio. Thus, 1D NMR (T_1 or T_2 distribution) measurements can be used to estimate pore size distribution (e.g., Wang & Ni, 2003; Zhang et al., 2016), determine irreducible and movable liquid saturation by comparing the T_2-distribution of a fully saturated sample with that of a sample after centrifugation (e.g., Lai et al., 2018; Lyu et al., 2020), and estimate hydraulic conductivity via empirical relationships such as the Schlumberger Doll Research (SDR) model, which links hydraulic conductivity with porosity and T_2 under saturated conditions (e.g., Behroozmand et al., 2015; Lucas-Oliveira et al., 2020). Parameter κ provides insights into the mechanism of surface interactions affecting NMR relaxation behaviors. For unsaturated pore space, we also can utilize κ to reflect the effect of surface interactions on NMR responses (Falzone & Keating, 2016a). In Equation 4, l is governed by effective pore geometry size, which reflects the proportion of pore space occupied by water. In unsaturated conditions, the water-occupied space is smaller compared to that in saturated conditions. This reduction in effective pore geometry size, representing a limited l, results in a lower κ. Consequently, this leads to spins diffusing to relaxation sites on the surface over a shorter timescale in water-occupied space compared to the saturated pore space. The impact of surface interaction on unsaturated NMR responses will be demonstrated in the following discussion regarding the unsaturated relaxation behaviors of single spherical pore.

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Two-dimensional (2D) NMR measurements can provide complementary information to 1D NMR measurements (e.g., Butler et al., 1981; Tan et al., 2012; Venkataramanan et al., 2002). The T_1-T_2 map is a typical 2D NMR output. When B_1 is reapplied to disrupt the relaxation process at a specific point, the transverse relaxation flowing the removal and subsequent reapplication of B_1 can be expressed as:

$$M_z(t_i, t_j) = M_0 \sum_j \sum_i f_{ji} \left(1 - e^{-\frac{t_i}{T_{1i}}}\right) \cdot e^{-\frac{t_j}{T_{2j}}} \tag{6}$$

In this context, f_{ji} represents the fraction of magnetization relaxing longitudinally with time constant T_{1i} and transverse relaxation time T_{2i}. Here, t_i is the time at which B_1 is reapplied, while t_j refers to any subsequent time point following the reapplication of B_1. By varying the timing of B_1 reapplication, the values of T_{1i} and T_{2j} can be plotted against f_{ji}, creating the 2D T_1-T_2 map. The contrast in T_1/T_2 ratios originates from the different intra- and inter-molecular NMR interactions occurring in various fluids (Fleury & Romero-Sarmiento, 2016), such as non-aqueous phase fluids with differences between T_1 and T_2 relaxation time. These interactions can be influenced by the mineral composition of clay (Fleury et al., 2022), rock materials (Straley, 1997), and other fluids within the pore environment (e,g. adsorbed gas in nanoporous materials at 100 K increases the T_1/T_2 ratio of methane, as noted by Riehl & Koch, 1972). The T_1-T_2 map is therefore useful for distinguishing between different fluid types (e.g., water, oil, methane, and other non-aqueous phase fluid) based on their distinct T_1/T_2 ratios (Daigle et al., 2014).

3. Development of the Simulation Framework of NMR Responses in Porous Media

3.1. Description of Magnetization Evolution in the Pore Space

The scenario of confined unsaturated pore space is shown in Figure 2a. Ω_1 and Ω_2 represent fluid- (e.g., water and oil) and air-occupied spaces in the pore space respectively, and Γ_1 and Γ_2 are boundaries that represent the pore's inner wall and air-fluid interface respectively. In the simulation framework, the governing equation is used to describe the magnetization vector of Ω_1 representing fluid over time and space, and boundary conditions qualify the magnetization at boundaries represented by Γ_1 and Γ_2.

The evolution of magnetization during excitation and the subsequent NMR relaxation of the energized nuclei in the fluid retained in pore space can be described by the Bloch-Torrey equation (Bloch, 1946; Torrey, 1956):

$$\frac{\partial \mathbf{m}}{\partial t} = \gamma(\mathbf{m} \times \mathbf{B}^{\mathrm{ex}}) - \begin{Bmatrix} m_x/T_{2\mathrm{B}} \\ m_y/T_{2\mathrm{B}} \\ (m_z - m_0)/T_{1\mathrm{B}} \end{Bmatrix} - \nabla \cdot (D\nabla \mathbf{m}) \tag{7}$$

where $\mathbf{m} = [m_x, m_y, m_z]^{\mathrm{T}}$ is the magnetization density vector, and γ is the gyromagnetic ratio of the protons. $\mathbf{B}^{\mathrm{ex}} = [B_x, B_y, \Delta\omega_1/\gamma + \Delta\omega_2(x, y, z)/\gamma]^{\mathrm{T}}$ is the external magnetic field in addition to $\mathbf{B}_0 = [0, 0, B_0]^{\mathrm{T}} = [0, 0, \omega_{\mathrm{L}}/2\pi\gamma]^{\mathrm{T}}$, where ω_{L} is the angular Larmor frequency of procession along the z-axis, and $\mathbf{B}_1 = [B_x, B_y, 0]^{\mathrm{T}}$ is the radio frequency (rf) field of the NMR excitation pulse. $\Delta\omega_1$ denotes the frequency offset due to chemical shift caused by protons in different chemical environments. $\Delta\omega_2$ represents the spatial frequency offset due to magnetic field inhomogeneities (local gradients) in the static field $\mathbf{B}_0$, which results in signal dephasing of the magnetization in the observational xy-plane. Because we assume an ideal external environment with a homogeneous magnetic field, $\Delta\omega_2$ is neglected in the following simulation. $T_{1\mathrm{B}}$ and $T_{2\mathrm{B}}$ are bulk relaxation time along the longitudinal and transverse directions. The diffusion tensor $\mathbf{D}$ simplifies to $D = \|\mathbf{D}\|$ in the case of isotropic diffusion being assumed in this study. As shown in Figure 2a, the Bloch-Torrey equation describes the magnetization vector of Ω_1 representing fluid (e.g., water and oil) over time and space.

In the confined pore space, the relaxation rate of excited nuclei is further accelerated due to surface interactions at solid-fluid interfaces. This surface relaxation mechanism is strongly dependent on the magnetic dipolar interactions of the spins in the fluid with magnetic dipoles at the fluid-solid interface (Foley et al., 1996; Torrey, 1956). The fluid-solid surface is represented by Γ_1 and defined as an active boundary with the boundary condition as follows:

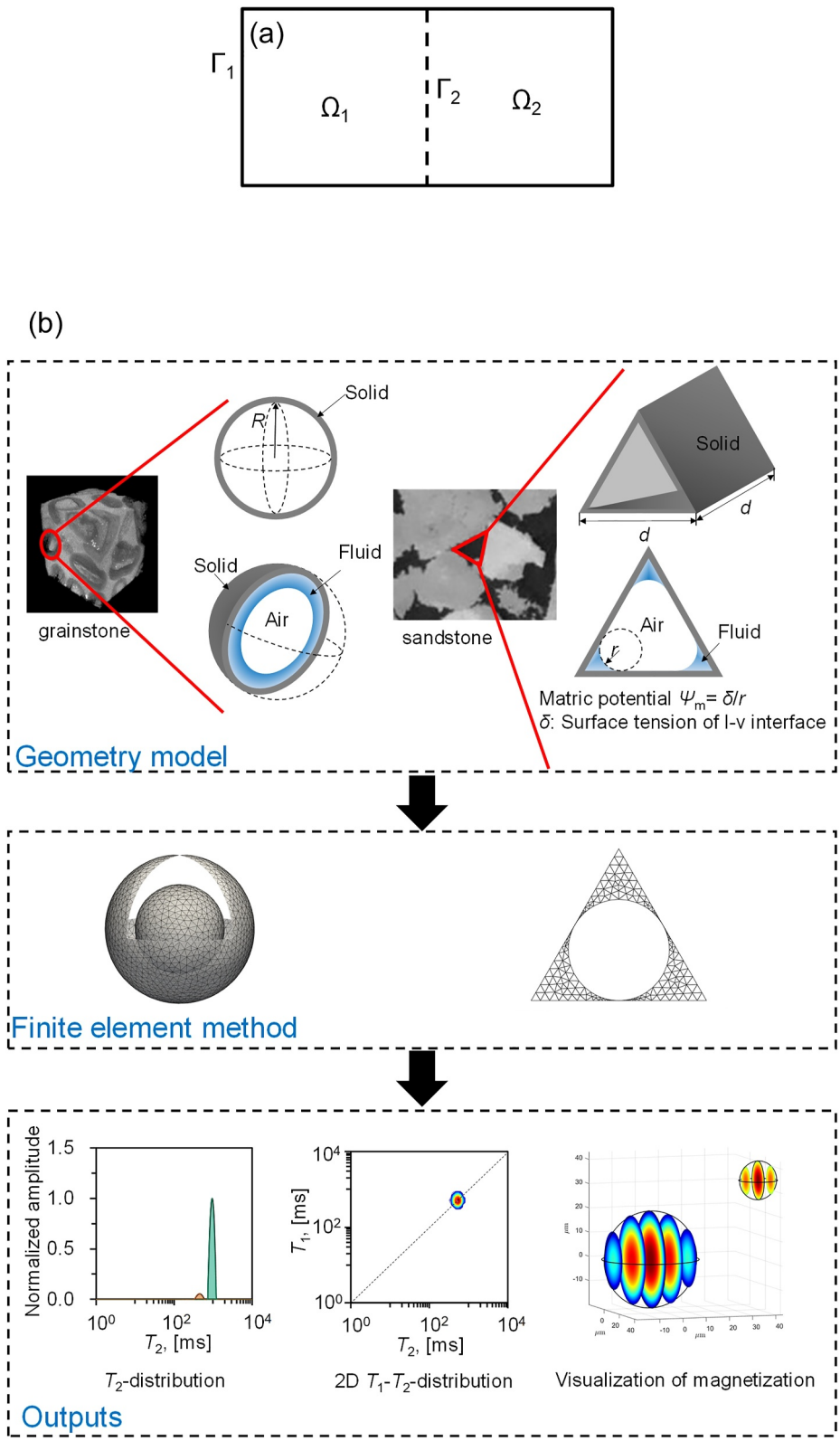


Figure 2. Description of the simulation framework of NMR response: (a) Scenario of confined unsaturated pore space; (b) The typical workflow for simulation of the NMR relaxation process in porous media.

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$$D\mathbf{n}\cdot\nabla\mathbf{m}+\rho_s\mathbf{m}\big|_{\Gamma_1}=0 \tag{8}$$

where $\mathbf{n}$ is the unit normal vector of boundary, and ρ_s is surface relaxivity. In the unsaturated system, as the air is a noncontributing passive volume (the magnetization vector in the air within Ω_2) (Mohnke & Klitzsch, 2010), the air-fluid interface Γ_2 can be treated as a passive boundary, which can be written as:

$$D\mathbf{n}\cdot\nabla\mathbf{m}=0 \tag{9}$$

Equations 8 and 9 are boundary conditions that are used to qualify the magnetization at fluid-solid and air-fluid interface respectively.

Collectively, Equations 7–9 can be combined to present the evolution of magnetization during excitation and the subsequent NMR relaxation of the energized nuclei in a single fluid phase trapped in porous media under the unsaturated state. Therefore, the NMR response of unsaturated to saturated porous media can be simulated by solving these equations using the finite element method with a commercial finite element software.

3.2. Numerical Simulation Process of NMR Responses

Figure 2 presents the workflow of numerical simulation of NMR responses in this study. To simulate the NMR relaxation process, we first establish the geometry model to represent the pore in the media. While the pore shape and pore size distribution are extremely complex in natural soil or rock, we use the idealized simple geometry model in our simulation. As shown in the CT image of grainstone and sandstone in Figure 2b, the pore space inside the solid matrix or between solid particles can be idealized as the spherical pore and triangle-shaped pore, respectively. In this study, we use spherical and triangle-shaped pore models to represent the unit cell of porous media, while the pore size distribution of porous media can be represented by constructing multiple pores with different sizes. The variations in fluid content and distribution within the media at different saturation stages can also be demonstrated using different thicknesses of water films and varying areas enveloped by the liquid-vapor interface.

Simulation can provide the spatial distribution of magnetization and allow convenient visualization of the time-dependent diffusion process while reducing computational costs and uncertainties by solving the diffusion equations in the eigenvalue domain to directly reflect one specific pore space in the media. The different NMR outputs - T_2-distribution and 2D T_1-T_2 map - can also be simulated by applying different time-dependent external magnetic field variations (see details in Appendix A and B).

4. Correction of Misinterpretation From NMR Responses of Unsaturated Porous Media

In this section, we will use the developed simulation framework of NMR response to correct the aforementioned misinterpretation of unsaturated water distribution dynamics. We propose an explanation for this misinterpretation: unsaturated pores can display short relaxation times akin to saturated smaller pores. This explanation demystifies the phenomena of the increased amplitude of shorter relaxation time components in T_2-distributions of unsaturated non-expansive, multi-pores media compared to their saturated state. To verify this explanation, we utilize a single spherical pore model to present the unsaturated pore water dynamic process and its NMR responses. Then, we use two spherical pores with different sizes as the unit cell to construct the dual-pore system, simulating the unsaturated pore-water dynamics of multi-pore media and its NMR responses. The simulated NMR responses will elucidate the reasons behind the misinterpretation and provide the correct interpretation of the unsaturated water distribution dynamics.

4.1. Simulation of T_2-Distribution in Unsaturated Single Spherical Pore

We establish the imbibition (or drainage) process of a single spherical pore to describe the pore-water dynamics. The imbibition (or drainage) in the pore space is achieved by the varying thickness of a thin film adhering to the pore's inner wall (e.g., Anand & Hirasaki, 2007), as shown in Figure 3a. During the imbibition, the water film forms and its thickness increases, eventually filling the pore space as it reaches saturation, representing the fluid distribution characteristic within unsaturated pore space. The drainage process of a single spherical pore is the

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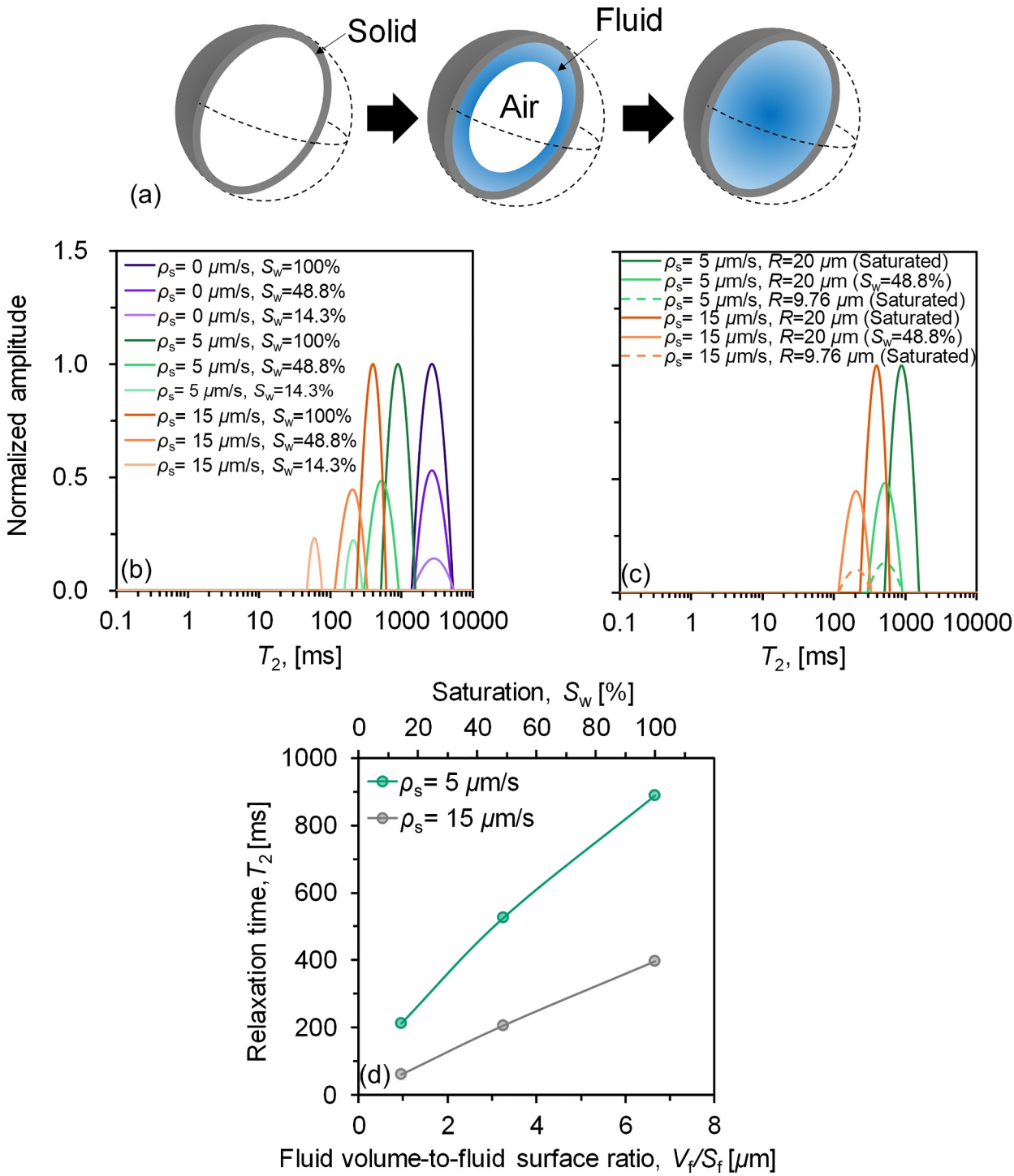


Figure 3. The imbibition process of the single spherical pore and its simulated NMR response: (a) the imbibition process of the single spherical pore; (b) the simulated T_2-distribution of the single spherical pore with different surface relaxivity under different saturation states; (c) the unsaturated spherical pore display short relaxation times akin to saturated smaller spherical pore; (d) the comparison of the variation of peak T_2 time and V_f/S_f of the single spherical pore during the imbibition process.

reverse of its imbibition process. In this section, we use the spherical pore with R = 20 μm as the pore radius for simulation. We set up three surface relaxivity ρ_s values as 0 μm/s, meaning the relaxation rate of excited nuclei is not accelerated by the surface interactions at the rock-fluid interface, and 5 and 15 μm/s, which is commonly seen for rock samples such as sandstone and limestone (e.g., Falzone & Keating, 2016a; Jiang et al., 2021; Zhang &

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Zhang, 2021). Applying time-dependent external magnetic field variation to simulate the CPMG (Carr-Purcell-Meiboom-Gill) pulse sequence (see detail in Appendix A), we utilize the developed simulation framework to simulate the evolution of T_2-distribution of the single spherical pore during the imbibition process.

As shown in Figure 3b, distinct surface relaxivity values cause different T_2 evolution during the imbibition process. The peak amplitude is normalized by dividing the maximum amplitude value of the peak during the imbibition process. From unsaturated to saturated state, in the absence of surface interactions, the T_2 amplitude increases, while the peak position remains unchanged. In this case, the surface relaxation term is removed in Equation 3, and the diffusion relaxation is also neglected within the ideal environment without internal inhomogeneous magnetic fields (Connolly et al., 2019) in our simulation. Equation 3, which can be written as $T_{1,2} = T_{1,2B}$, indicates NMR relaxation dominated by fluid bulk relaxation. This case occurs in porous media when pore spaces only contain non-wetting fluids (Al-Garadi et al., 2022; Freedman et al., 2003) or when extremely large pores (greater than the millimeter scale) are fully saturated with the water. T_2 relaxation time peaks with varying amplitude but unchanged positions do not provide information about pore structure, while the amplitude of peaks reflects the water content in pore space. However, for $\rho_s = 5$ and 15 μm/s, not only does the amplitude of the peak in T_2-distribution increase, but the peak also shifts from short to long relaxation time components. While $\rho_s = 5$ and 15 μm/s could provide additional insights, it does not alter the observed trend where the relaxation time peak shifts to shorter times as the pore space transitions from saturated to unsaturated states. Comparing the evolution of the T_2-distribution for $\rho_s = 0$ μm/s and $\rho_s = 5$ and 15 μm/s, Figure 3b reflects that surface interactions can accelerate the NMR relaxation process in unsaturated compared to saturated pore space. This suggests that surface relaxation induces the shift of the T_2 peak from short to long relaxation time components during the imbibition process. Figure 3c shows that, whether for $\rho_s = 5$ and 15 μm/s, the unsaturated pore with $R = 20$ μm can display short T_2 relaxation times similar to a saturated pore with $R = 9.76$ μm. Both our simulated results in Figures 3b and 3c confirm that unsaturated pores exhibit short relaxation times similar to those of saturated but smaller pores due to surface interactions at solid-fluid interfaces accelerating the NMR relaxation process. This reflects the vital role of the surface relaxation mechanism in NMR relaxation behaviors within pore space during imbibition.

In the unsaturated compared to the saturated pore space, the l in Equation 4 is limited, resulting in a smaller parameter κ, leading to surface relaxing spins diffusing throughout the water-occupied space over a shorter timescale. To demonstrate the effect of surface relaxation on NMR relaxation behaviors of the unsaturated pore system, similar to Equation 5, we utilize the fluid volume in the pore space (V_f) and the surface area between fluid and solid (S_f) to define the generalized fluid volume-to-fluid surface area ratio (V_f/S_f). For the single spherical pore, the comparison of variation of V_f/S_f and peak T_2 value during the imbibition process is shown in Figure 3d. Whether for $\rho_s = 5$ and 15 μm/s, the V_f/S_f ratio increases due to the rise of saturation, and the peak T_2 value increases with the increase of V_f/S_f and saturation during imbibition, suggesting that as saturation increases, the water-occupied space expands, requiring spins to diffuse over greater distances to reach surface relaxation sites. This results in fewer interactions between protons and the surface, making surface interactions less effective at enhancing the NMR relaxation throughout the water-occupied space. Consequently, this leads to a shift in the relaxation time peak from unsaturated to saturated pore space. This definition and understanding of the V_f/S_f ratio is adopted in the following discussion regarding the different unsaturated relaxation behaviors in various pore shapes and pore sizes.

4.2. Simulation of T_2-Distribution in Unsaturated Dual-Spherical Pore System

Following the single pore system, in this section, we simulate the dual-pore system, which is constructed from two spherical pores of different sizes. This simulation aims to model the pore-water dynamics within a multi-pore system and analyze its T_2-distribution. The drainage or imbibition process usually presents the unsaturated pore-water dynamics. In many rocks and sands, capillarity, arising from liquid-vapor interactions due to the cohesive forces among water molecules, generally dominates across a wide range of unsaturated water content during the water dynamics process. The higher capillary pressure in smaller pores creates a pressure differential that drives crossflow between small and large pores (Wang et al., 2023). During drainage, the wetting phase drains from the largest pores first, while during imbibition, capillary suction causes the wetting phase to imbibe into the smallest pores first (Kovscek et al., 1993). Several studies' observations (e.g., Lehmann & Or, 2009; Wang et al., 2023) and simulation results (e.g., Zheng et al., 2018) also support that the water drains from large pores first during drainage and imbibes into small pore spaces first during imbibition. As a result, the contribution of

small and large pores to total water content remains similar during both drainage and imbibition. The water dynamic distribution and observed trend of unsaturated NMR responses (e.g., T_2-distributions of synthetic sands during the drainage-imbibition process in Falzone & Keating, 2016b) are not significantly different during the drainage and imbibition. In this work, we will use imbibition, during which water imbibes into the small pore first, to present the unsaturated pore water dynamic, studying the unsaturated NMR responses of the dual-pore system. We set up two spherical pores with the radius of $R_1 = 20$ μm, and $R_2 = 8$ μm to represent multi-pore media. Because the non-expansive natural porous media, such as sandy soil and sandstone, exhibits mineralogical homogeneity, the surface relaxivity of different pore environments within the media is not distinct. To accurately simulate the unsaturated NMR responses of the media using the dual-pore system, we set the surface relaxivity of both the large and small pore in the system to $\rho_s = 5$ μm/s. For $\rho_s = 5$ μm/s, because the κ of spherical pore with $R = 20$ μm is $0.05 < 0.1$, the system remains within fast regime. Similar to the single pore system, the imbibition process is simulated to present the different saturation states in the dual-pore system. Due to the higher imbibition rate of small pores than that of large pores during the imbibition process (e.g., Wang et al., 2023; Yang et al., 2016; Zheng et al., 2018), the small spherical pore is considered to be filled first until it reaches a saturated state, followed by the large pore reaching saturation (Figure 4a). We utilize the simulated evolution of T_2-distribution during the imbibition process to recreate the misinterpretation and unveil the mechanism resulting in the misinterpretation.

The simulated T_2-distribution evolution of the dual-pore system during the imbibition process is presented in Figure 4a. Saturation S_w is the proportion of water volume V_w in the total volume of the pore space V_{total}, $S_w = V_w/V_{total}$. During the imbibition, the T_2-distribution initially presents only one relaxation time peak, which reflects water in the small pore before it reaches saturation, as observed at $S_w = 3.48\%$. Once the small pore is saturated, such as at $S_w = 25.6\%$, the T_2-distribution shows two relaxation peaks, representing water in both the small and large pore. Compared with the system at $S_w = 28\%$, the system at $S_w = 44.3\%$, 79.7%, and 100% shows a decrease in the amplitude of shorter relaxation T_2 components. As shown in Figure 4b, the varying component saturation, which is the ratio of water volume represented by different relaxation time peaks to total pores volume, directly demonstrates the misinterpretation of the water dynamic distribution in the multi-pore system from the T_2-distribution. At $S_w = 28\%$, if a shorter relaxation time peak is used to represent the small pore, it would erroneously imply that the small pore exceeds its saturation water capacity, which is physically impossible. This illustrates the specific misinterpretation in the analysis of unsaturated water dynamic distribution within the multi-pore system based on the assumption that Equation 5 remains valid.

To address this misinterpretation, we simulate the T_2 peak evolution of each pore during the imbibition process, providing a more accurate representation of the water distribution dynamics in the multi-pore system. In Figure 4c, the green peak represents the large pore and the orange peak represents the small pore. After the small pore is saturated ($S_w = 28\%$), the shorter relaxation time peak primarily corresponds to water in the large pore. At $S_w = 44.3\%$, the two relaxation time peaks overlap. At $S_w = 79.7\%$ and 100%, the longer relaxation time peak represents the water in the large pore and the shorter relaxation time represents the water in the small pore. Figure 4c demonstrates the correct interpretation of water dynamic distribution, in which the small pore does not exceed its saturation capacity.

The variation of V_f/S_f of the large and small pore during the imbibition process reveals that the effect of surface relaxation leads to the misinterpretation of unsaturated water dynamic distribution. At $S_w = 28\%$, the smaller V_f/S_f of the large pore implies enhanced NMR relaxation due to surface relaxation in the unsaturated large pore compared to the water-filled small pore, resulting in a shorter relaxation time for the large pore. At $S_w = 44.3\%$, equal V_f/S_f values indicate identical relaxation times for both the small and large pore, causing the two peaks to overlap. At $S_w = 79.7\%$ and 100%, the greater V_f/S_f of the large pore indicates a longer relaxation time peak representing the large pore. The variation of V_f/S_f during imbibition reflects the different evolution of NMR relaxation influenced by surface relaxation in pores of different sizes. At low saturation, the V_f/S_f values of the large pore are not greater than that of the small pore, implying that the shorter relaxation time peak of the large pore or the overlap of relaxation times peaks of the two pores induces the potential misinterpretation of water dynamic distribution from the T_2-distribution of porous media.

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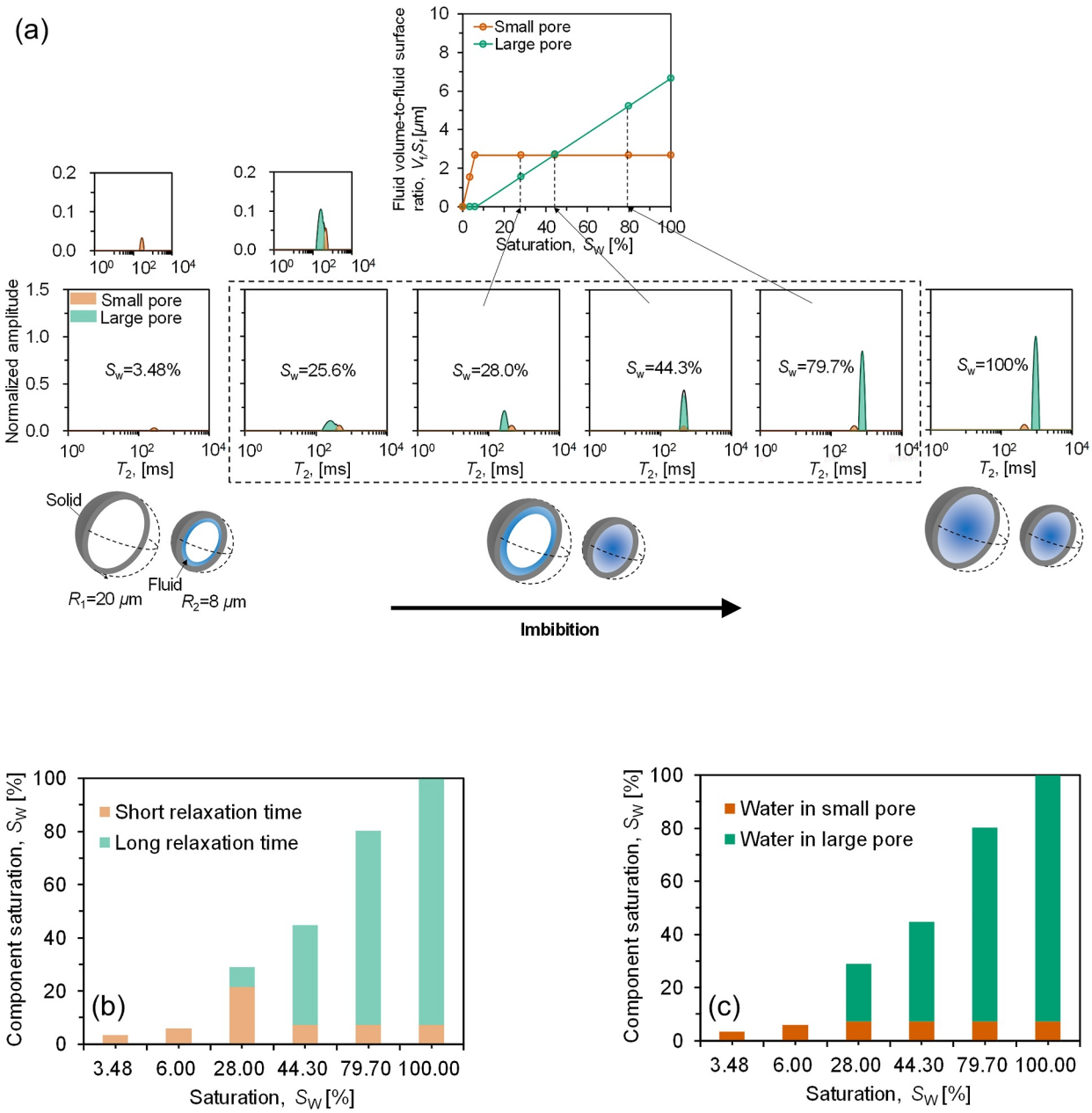


Figure 4. (a) The simulated evolution of T_2-distribution and variation of V_f/S_f of dual-spherical pore system during the imbibition process; (b) the water partitioning of the dual-spherical pore system revealed by different relaxation time peaks; (c) the pore water dynamic distribution of the dual-spherical pore system.

5. The Effect of Pore Structure Characteristics on Unsaturated NMR Relaxation Behaviors

In the previous section, we focus on the effect of pore sizes on the unsaturated NMR relaxation behaviors through surface relaxation. In this section, we use the simulation framework to investigate how pore structure (pore size distribution and shape) affects the NMR relaxation in the unsaturated dual-pore system.

5.1. Simulation of T_2-Distribution in Unsaturated Dual-Triangular-Shaped Pore System

Instead of the spherical pore, we set up the triangular-shaped pore as the unit cell to construct the dual-pore system. We have two equilateral triangular-shaped pores with side lengths of d_1 = 20 μm and d_2 = 8 μm and

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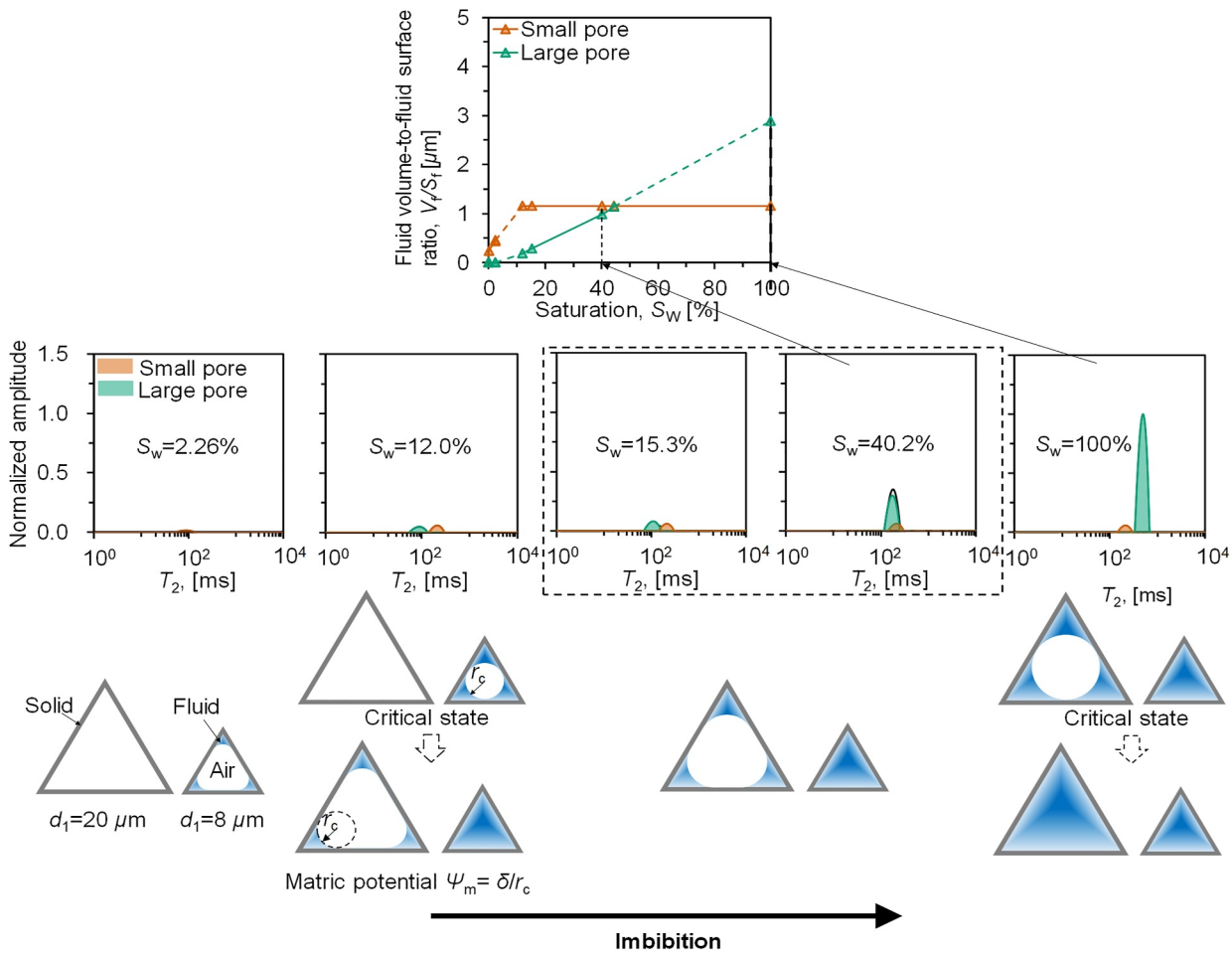


Figure 5. The simulated evolution of T_2-distribution and variation of V_f/S_f of dual-triangle-shaped pore system during the imbibition process.

we also set surface relaxivity for $\rho_s = 5$ μm/s. For $\rho_s = 5$ μm/s, because the κ of equilateral triangular-shaped pore with $d = 20$ μm is $0.0503 < 0.1$, this system also remains within fast regime. We point out that the imbibition process in a triangular-shaped pore differs from that in a spherical pore. As shown in Figure 2c, during imbibition, the radius of curvature r increases until the separate curved liquid-vapor interface, known as arc meniscus, in the corners come into contact, leading to spontaneous pore filling (referred to as "snap off"). The Young-Laplace equation (Young, 1805) describes the relationship between capillary pressure or matric potential and the radius of curvature r. The imbibition process of the triangle-shaped pore can be used to model hydrodynamic liquid behaviors in unsaturated porous media (Tuller et al., 1999). Mohnke et al. (2015) used equilateral and right-angled triangular pores to simulate unsaturated NMR responses represented by T_1-distribution, investigating the effect of pore shape, capillary pressure, and water distribution of unsaturated NMR relaxation behaviors. In our study, we build on this approach by establishing a dual-triangle-shaped pore system to simulate unsaturated pore-water dynamics and NMR responses, focusing on how pore size influences unsaturated NMR relaxation behavior. Additionally, we explore the effect of pore shape on unsaturated NMR relaxation signals by comparing simulated NMR responses of the dual-triangle-shaped and dual-spherical pore system. As shown in Figure 5, to describe the higher imbibition rate of small pores compared to large pores during imbibition, an arc meniscus forms in the small pore first. The radius of curvature of this arc meniscus r_1 increases until the separate arc meniscus of the corners touch ($r_1 = r_c$, reaching a critical state), and the pore is saturated spontaneously. Subsequently, the arc meniscus forms in the large pores with the radius of curvature $r_2 = r_c$ corresponding to the matric potential at that moment. As the large pore continues to be filled, the r increases until the separate arc menisci of the corners touch, reaching a critical state, and the pore is saturated spontaneously.

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The evolution of the T_2-distribution of the dual-triangular pore system and the V_f/S_f of the large and small pore during imbibition are presented in Figure 5. Even though the variation of V_f/S_f and the evolution of T_2-distribution of the dual-triangular pore system broadly resemble those in the dual-spherical pore system during imbibition, there are notable differences in NMR relaxation behaviors between the two systems. These differences are reflected in the variation of V_f/S_f and the evolution of T_2-distribution. Due to the presence of two critical states during the imbibition process in the dual-triangular pore system, the V_f/S_f of the small pore can increase spontaneously at the first critical state, while the V_f/S_f of the large pore increases spontaneously at both critical states. Notably, before the large pore reaches the critical state, the V_f/S_f of the large pore is smaller than that of the small pore. This implies that the two relaxation peaks never overlap, and the shorter relaxation time peak always represents the water in the large pore until the system is fully saturated.

5.2. The Effect of Pore Shape and Size Distribution on Unsaturated NMR Relaxation Behaviors

The variation of V_f/S_f reflects the evolution of NMR relaxation behaviors affected by surface relaxation in unsaturated pore-water dynamics. Equal V_f/S_f values for the large and small pore indicate identical T_2 relaxation times and the overlap of relaxation time peaks of the two pores. When the system saturation is lower than the saturation corresponding to equal V_f/S_f, the surface relaxation causes the unsaturated large pore to have a shorter relaxation time than the saturated small pore. In this section, we utilize the saturation point of equal V_f/S_f to investigate the effect of pore shape and size distribution on the NMR relaxation behaviors of unsaturated porous media.

Figure 6 shows the variation of the V_f/S_f in dual-spherical and dual-triangular pore systems with different pore size distributions during imbibition. Different pore size distributions are achieved by varying R_2 and d_2. The comparison of V_f/S_f variations in Figures 6a, 6c, and 6e with those in Figures 6b, 6d, and 6f reflects the effect of pore shape on NMR relaxation behaviors through surface relaxation. Even with similar pore size distributions, the saturations of equal V_f/S_f in dual-triangular pore systems are lower than those in dual-spherical pore systems. Specifically, as shown in Figure 6f, the saturation of equal V_f/S_f in the dual-triangular pore system with d_2 of 4 μm is 13.4%, which is significantly lower than 21.5% observed in the dual-spherical pore system with R_2 of 4 μm. Utilizing equal V_f/S_f to represent the overlap of two relaxation time peaks, the evolution of the T_2-distribution can reflect the pore shape within rock or soil. Our results suggest that for two rock or soil samples with similar pore size distributions, if the pore shapes in the two samples are different, even if their T_2-distributions exhibit distinct two relaxation time peaks with similar characteristics (e.g., amplitude and position) under the saturated state, the saturations of the overlap of two relaxation time peaks (i.e., equal V_f/S_f) in their T_2-distribution are distinct under the unsaturated state. If the overlap of two peaks in T_2-distribution of one sample occurs at lower saturation, the pore shape within this sample tends to be more triangular, while the pore shape in the other sample is more spherical.

The different variations of V_f/S_f shown in Figures 6a, 6c, and 6e reflect the effect of pore size distribution on NMR relaxation behaviors in unsaturated porous media. Whether constructed by spherical or triangular pores, the saturation of equal V_f/S_f decreases as the small pore size decreases, leading to a lower saturation at which the overlap of two relaxation time peaks occurs. The decrease in the small pore size would induce the misinterpretation of water dynamic distribution from the T_2-distribution of porous media at lower saturation, implying that the effect of pore size distribution on NMR relaxation behaviors should be considered in the interpretation of T_2-distribution in unsaturated pore-water dynamics.

This study, incorporating the simulated NMR responses of dual-pore systems with varying size distributions and shapes, highlights the significant effects of different pore sizes and shapes on unsaturated NMR relaxation behaviors. In summary, different pore shapes and pore size distributions induce different water retention behaviors in multi-pore systems, leading to distinct unsaturated NMR relaxation behaviors due to surface relaxation. Therefore, considering pore structure characteristics is crucial when interpreting NMR responses in unsaturated pore-water dynamics.

6. Numerical Simulation of 2D T_1-T_2 Map Evolution in Multi-Pore System

Our developed simulation framework, applying time-dependent external magnetic field variations to simulate saturation recovery (SR)-CPMG sequence pulse (details in Appendix B), also can be used to simulate the 2D T_1-T_2 map of a muti-pore system. In this section, we consider two spherical pores with radii of $R_1 = 20$ μm and

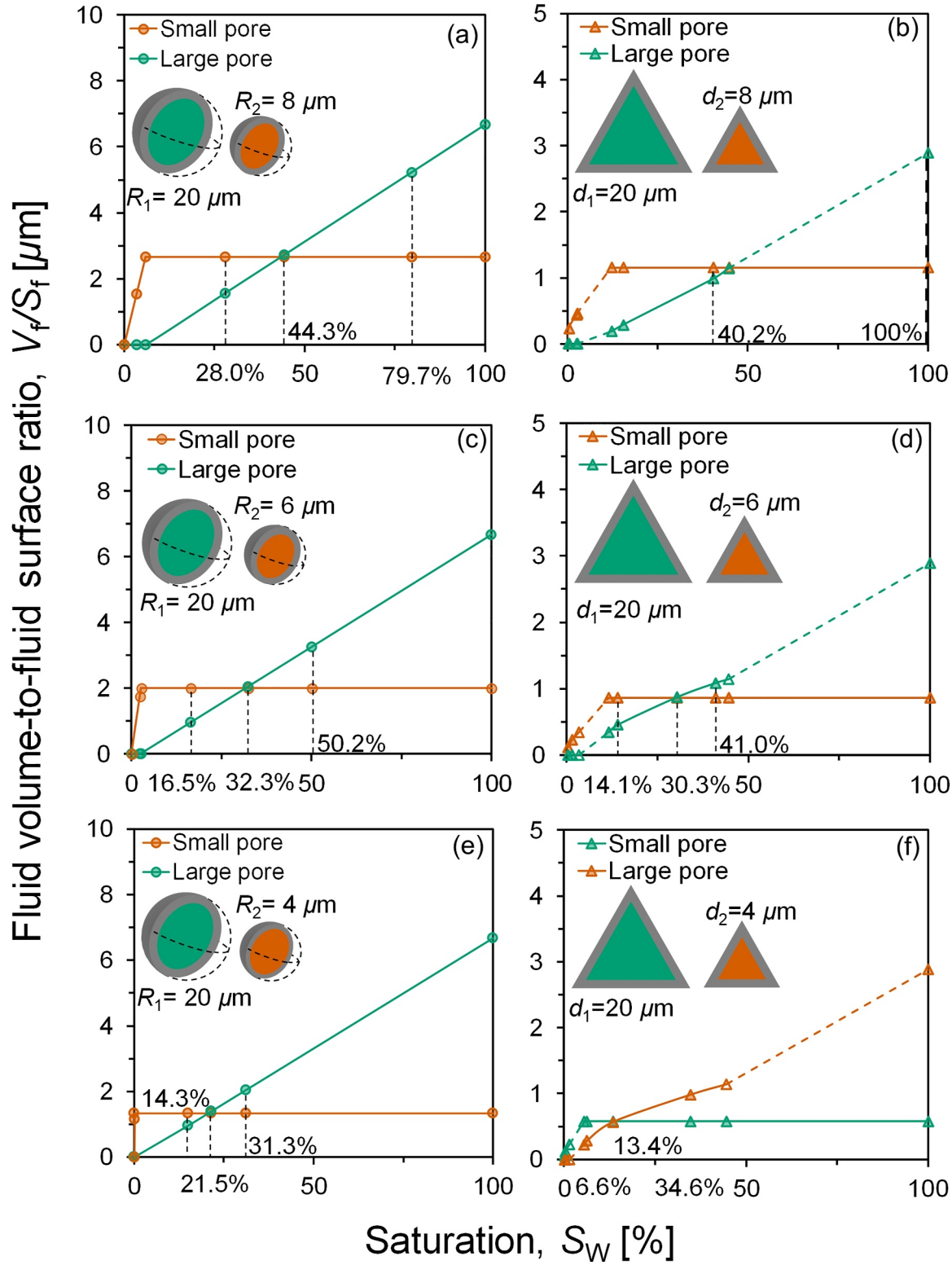


Figure 6.

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$R_2 = 8$ μm, and two equilateral triangular pores with side lengths of $d_1 = 20$ μm and $d_2 = 8$ μm. Similar to the previous simulation, we set up surface relaxivity ρ_s as 5 μm/s. The simulated evolution of the T_1-T_2 map of the dual-spherical and dual-triangular pore system during imbibition will elucidate the impact of pore size and shape on unsaturated NMR relaxation behaviors.

Figure 7 presents the evolutions of the T_1-T_2 map of the two dual-pore systems during imbibition after small pores reach saturation. The amplitude of the T_1-T_2 peak is normalized by dividing the peak amplitude by the maximum amplitude value at the respective saturation level during imbibition. The T_1-T_2 map evolutions of both the dual-spherical and dual-triangular pore system are broadly similar. For both systems, once the small pore is saturated, a single peak, representing the saturated small pore, is located on the $T_1/T_2 = 1$ line. As the large pore continues to be filled until the system is fully saturated, a peak representing water in the large pore shifts from below to above the peak of the saturated small pore along the $T_1/T_2 = 1$ line. Enhanced NMR relaxation in the unsaturated large pore results in its T_1-T_2 peak appearing below the peak of the saturated small pore along the $T_1/T_2 = 1$. In such an ideal pore environment without inhomogeneous magnetic fields, diffusion relaxation does not occur (diffusion relaxation does not impact T_2 relaxation anymore), and both T_1-T_2 peaks of the large and small pore shift along the line $T_1/T_2 = 1$ during the pore water dynamic process, implying not only the decay of transverse (T_2) magnetization but also the recovery of longitudinal (T_1) magnetization enhanced by the same surface relaxation. These evolutions of the T_1-T_2 map also imply that the various water dynamics in pores of different sizes, influenced by surface relaxation, result in the distinct evolution of NMR relaxation behaviors between small and large pores of porous material from the unsaturated to the saturated state. As shown in Figure 7a, the T_1-T_2 peaks representing water in the large and small spherical pore overlap at $S_w = 44.3\%$. In contrast, Figure 7b shows the T_1-T_2 peaks for the large and small triangular pores overlapping at $S_w = 40.2\%$. This difference also highlights the effect of pore shape on NMR relaxation.

Previous studies have utilized 2D T_1-T_2 map to study the variation of water components, such as movable (e.g., Jaeger et al., 2010; Tian & Wei, 2020) and adsorption water (e.g., Peng et al., 2023), in unsaturated rock or soil. However, no study has demonstrated the complicated pore environments impacting the interpretation of 2D T_1-T_2 maps. We utilize the developed simulation framework to clarify the effect of pore shape and size on not only the transverse relaxation but also the longitudinal relaxation. This study provides insight into interpreting the 2D T_1-T_2 map of soil and rock with complex pore structures.

The T_1-T_2 map allows us to use T_1/T_2 contrast to differentiate fluid components within porous media. In future work, we will extend this simulation framework to model the T_1-T_2 map of porous media containing multi-phase fluids, such as rock with pure water and saline solution, and sandstone with pure water and oil or other non-aqueous phase fluid. The simulated T_1-T_2 map will be used to analyze the saturation or content and distribution of fluids within natural porous media.

7. Correct the Interpretation of Experimental T_2-Distribution in Unsaturated Media

Many studies use NMR measurements to determine the water distribution in the different pore environments within saturated and unsaturated porous media. By fitting the T_2 distribution curve using the multi-Gaussian model, different Gaussian peaks can represent water-saturated pores of various sizes within the rock or soil. Such peak decomposition using multi-Gaussian fitting retrieves the peak position, width, and area, and has also been used for determining the pore size distribution of saturated rock or soil (e.g., Ge et al., 2022), to identify different fluids (e.g., oil and water) within reservoirs (e.g., Zhong et al., 2020). In unsaturated media, Zhang and Zhang (2021) and Peng et al. (2023) use different Gaussian peaks to represent water-saturated pores of distinct sizes within carbonate rock (Zhang & Zhang, 2021) or clay samples (Peng et al., 2023). However, the representation of each Gaussian component and the evolution of each peak have not been clarified. In Zhang and Zhang (2021) and Peng et al. (2023), the shorter and longer relaxation time Gaussian peaks obtained through peak decomposition do not accurately represent water in small and large pores, respectively, within their unsaturated

Figure 6. The variation of the V_f/S_f of the dual-spherical and -triangle-shaped pore systems with different pore size distributions during the imbibition process: (a) the variation of the V_f/S_f of the dual-spherical pore system with $R_1 = 20$ μm and $R_2 = 8$ μm; (b) the variation of the V_f/S_f of the -triangle-shaped system with $d_1 = 20$ μm and $d_2 = 8$ μm; (c) the variation of the V_f/S_f of the dual-spherical pore system with $R_1 = 20$ μm and $R_2 = 6$ μm; (d) the variation of the V_f/S_f of the -triangle-shaped system with $d_1 = 20$ μm and $d_2 = 6$ μm; (e) the variation of the V_f/S_f of the dual-spherical pore system with $R_1 = 20$ μm and $R_2 = 4$ μm; (f) the variation of the V_f/S_f of the -triangle-shaped system with $d_1 = 20$ μm and $d_2 = 4$ μm.

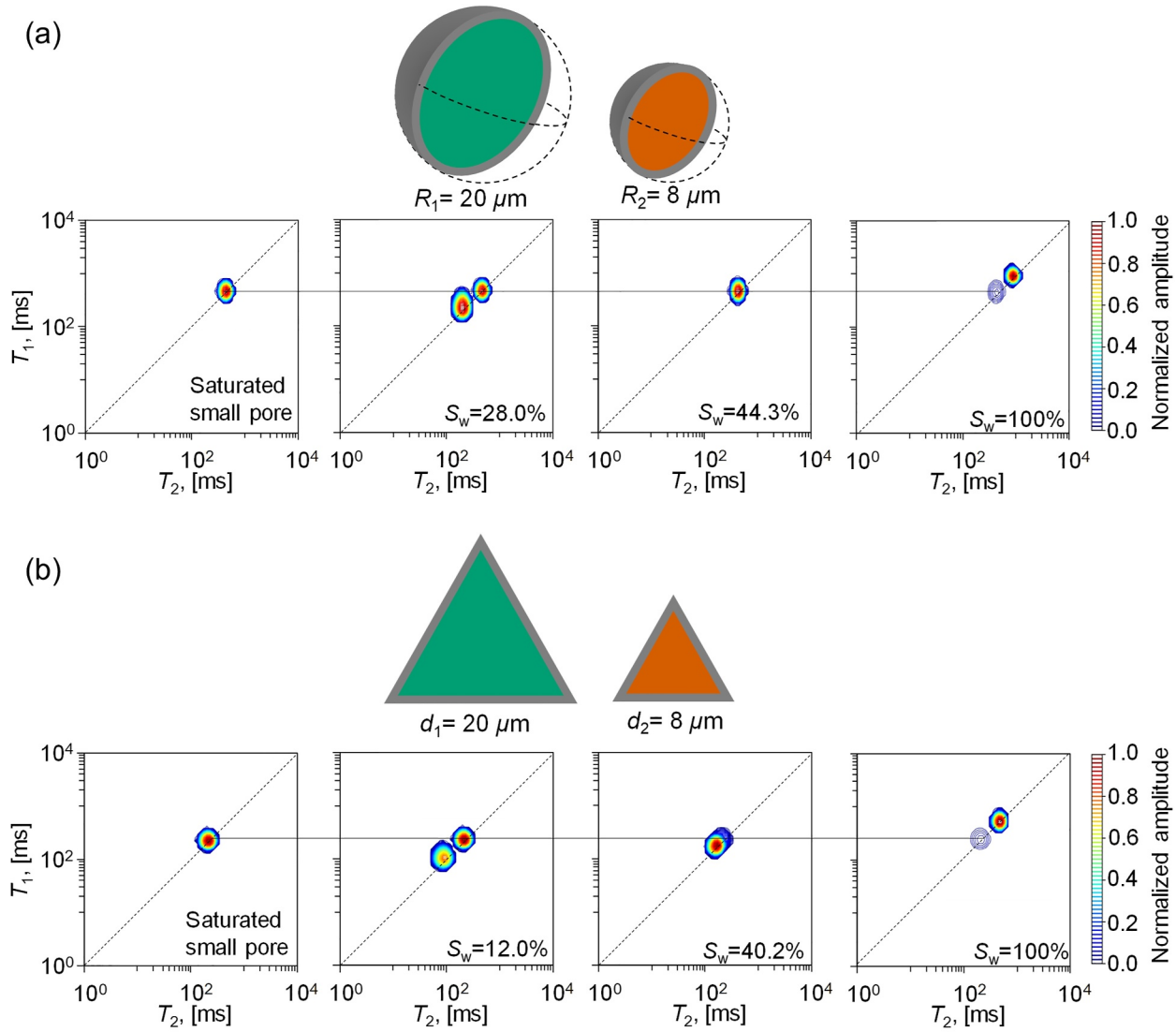


Figure 7. The simulated evolution of 2D T_1-T_2 map of (a) dual-spherical and (b)-triangular-shaped pore system during the imbibition process.

samples. Figure 8g shows water partitioning as interpreted and revealed by the peak decomposition method in Zhang and Zhang (2021). We also applied this peak decomposition method to an NMR data set on unsaturated silty sand to reflect water partitioning (Figure 8c). The direct findings from multi-Gaussian analysis incorrectly imply that small pores can exceed their saturation capacity, which is physically impossible. This specific misinterpretation was demonstrated in detail earlier in Section 4.

Here, we apply our method to interpret published experimental T_2-distribution data in unsaturated multi-pore media, revealing the true evolution of unsaturated pore water distributions. We consider two cases: Case 1 involves measurements on non-swelling silty sand reported by Meyer et al. (2018), and Case 2 is based on the study of Zhang and Zhang (2021) on unsaturated limestones.

Case 1 is an experimental study by Meyer et al. (2018), which uses silty sand (clay content: 4.7%; silt content: 12.7%; sand content: 82.6%) samples. In this case, matric potentials were adjusted stepwise after complete saturation of the soil samples with water to acquire samples at different saturations, and the NMR method was utilized to investigate the relationship between soil water content and matric potential by transferring amplitudes for relaxation time peaks into water content at different matric potentials. Figures 8a and 8b present the T_2-distributions of non-swelling silty sand at S_w = 100.0%, 29.1%, and 10.7%. At S_w = 100.0%, the T_2-distribution curve is decomposed into peak 1 and peak 2 to represent saturated small pores and saturated large pores,

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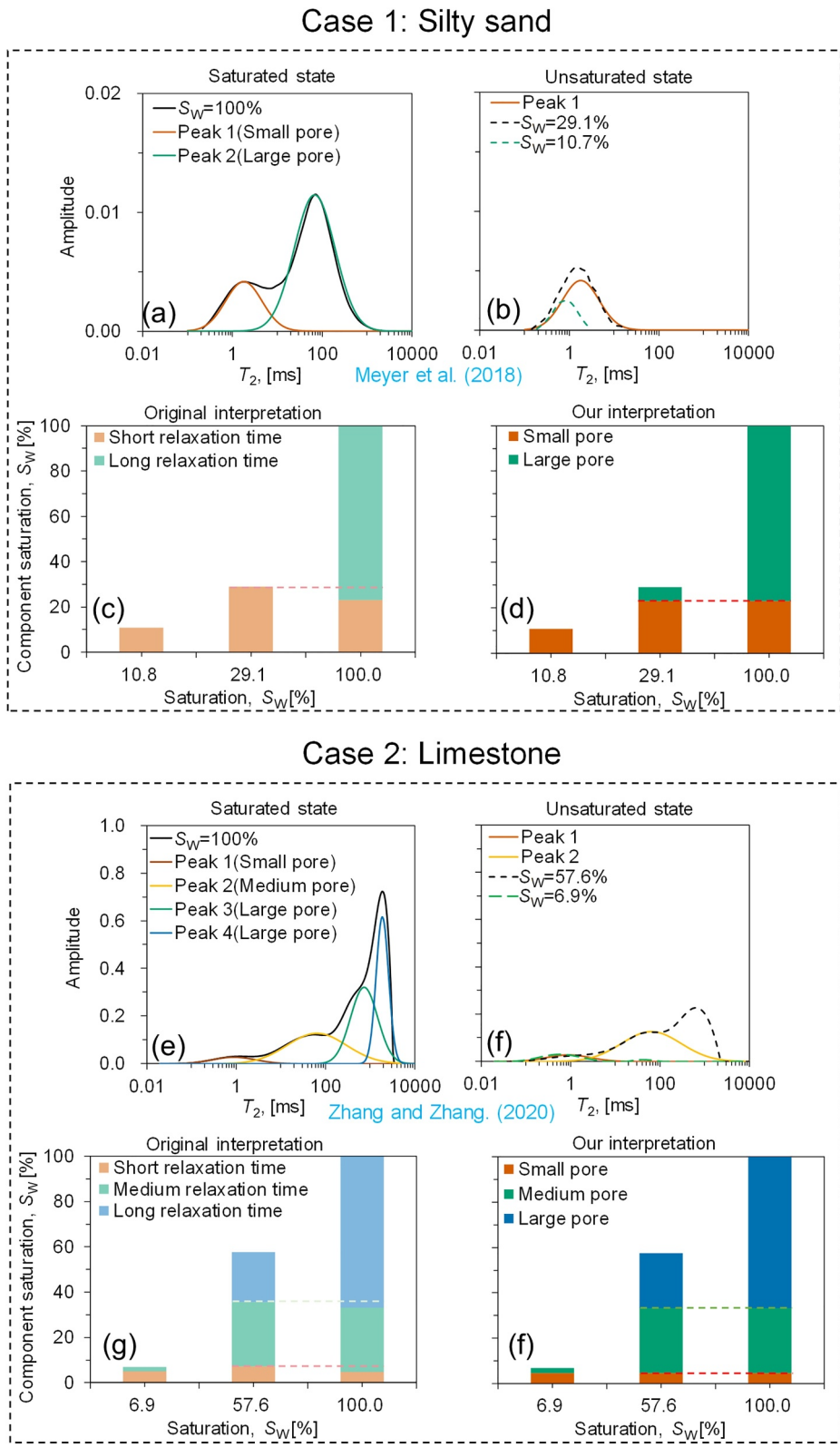


Figure 8. The unsaturated pore water dynamic distribution revealed by T_2-distribution: (a) Gaussian peak composition of saturated T_2-distribution curve and (b) unsaturated T_2-distribution curve of silty sand in Meyer et al. (2018); (c) water partitioning revealed by the peak composition method and (d) pore water dynamic distribution of silty sand; (e) Gaussian peak composition of saturated T_2-distribution curve and (f) unsaturated T_2-distribution curve of limestone in Zhang and Zhang. (2021); (c) water partitioning revealed by the peak composition method and (d) pore water dynamic distribution of limestone.

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respectively. The ratio of the area under peak 1 to the total T_2-distribution curve represents the contribution of large pores to saturation, while the ratio under peak 1 represents the contribution of small pores. At $S_w = 10.7\%$, the T_2-distribution curve is enveloped by peak 1, and its amplitude is smaller than peak 1, implying that small pores are unsaturated and large pores are not filled. The contribution of small pores to the saturation equals the saturation of this sample. At $S_w = 29.1\%$, the T_2-distribution curve clearly envelopes peak 1, indicating that small pores are saturated, and the increased amplitude of the T_2-distribution curve relative to peak 1 represents water in large pores. Clearly, the difference between the sample saturation and the contribution of saturated small pores presents large pores' contribution to saturation.

Case 2 is also an experimental study by Zhang and Zhang (2021), which uses limestone samples. In this case, saturated samples were dried in the chamber for different hours to obtain samples at different saturation levels, and NMR measurements were applied to quantify the dynamics of rock moisture partitioning during the drying process by decomposing T_2-distributions into multiple T_2 peaks representing water in different pore environments. Figures 8e and 8f demonstrate the T_2-distributions of limestone in Zhang and Zhang (2021) at $S_w = 100.0\%$, 57.6%, and 6.9%. At $S_w = 100.0\%$, the T_2-distribution curve is decomposed into four Gaussian peaks. Here we integrate peak 3 and peak 4 to represent saturated large pores and use peak 1 and 2 to represent saturated small and medium pores respectively. At $S_w = 6.9\%$, the T_2-distribution curve envelopes peak 4 but not peak 3, indicating that small pores are saturated and a small amount of water is retained in medium pores. Similar to Case 1, the contribution of small pores to saturation equals that of saturated small pores, and the difference between the sample saturation and this value represents the contribution of medium pores. At $S_w = 57.6\%$, the T_2-distribution curve envelopes peak 1 and 2, implying that small and medium pores are saturated, and the increased amplitude of the T_2-distribution curve relative to peak 1 and 2 represents water in large pores. Thus, the difference between the sample saturation and the enveloped area under peak 1 and 2 represents the contribution of large pores.

Figure 9 presents a flowchart outlining the use of T_2-distributions to determine the unsaturated water dynamics within porous media. Compared to the peak decomposition method used by Zhang and Zhang (2021), our interpretation method provides a more accurate description of pore-water dynamics. It ensures that the small pores do not exceed their maximum saturation capacity, avoiding misinterpretations. Our method can be widely applied to interpret unsaturated pore-water dynamics from the T_2-distribution in various natural porous media, thereby improving the accuracy and reliability of NMR data interpretation.

8. Broader Implications

Our enhanced NMR interpretation method addresses critical challenges in accurately analyzing water storage and distribution in unsaturated porous media, making it a valuable tool for researchers and practitioners in the field. Understanding water storage dynamics in the vadose zone is essential for managing water resources, predicting geological hazards, and mitigating environmental issues. Accurate data interpretation is vital for qualifying crucial processes such as runoff, evapotranspiration, and groundwater recharge.

Traditional NMR techniques often face limitations when applied to unsaturated conditions due to the complex architecture of soil and rock and fluid-rock interactions. Beyond pore size and shape discussed in this study, pore connectivity can also influence NMR experimental data. The phenomenon, known as pore coupling, can distort NMR relaxation time distributions when magnetization exchange occurs between connected pore environments during the NMR measurement period (see discussion in Soto Bravo & Zhang, 2023), leading to an averaged representation of the pore network, rather than capturing the characteristics of individual pore environments. The distinct mineralogy of solid matrices forming different pore environments in natural porous media leads to complex liquid-solid interactions, significantly influencing their unsaturated NMR responses. For example, in many natural porous media composed of particles of various sizes (e.g., sand, silt, clay, and gravel), the mineralogical differences between these particles lead to a contrast in pore surface relaxivity within the media. Macro-pores formed by sand or quartz exhibit low surface relaxivity, while micro-pores with clay as the solid matrix exhibit high surface relaxivity (e.g., Elsayed et al., 2023; Livo et al., 2020), inducing the overlap of relaxation time peaks between macro- and micro-pores at lower saturation, compared to the case in this study where the surface relaxivities of macro- and micro-pores are not distinct. Assuming the homogeneous pore surface relaxivity within the media may lead to further misinterpretations of unsaturated water distribution dynamics from NMR responses. Our study provides a robust framework that addresses these limitations, analyzes

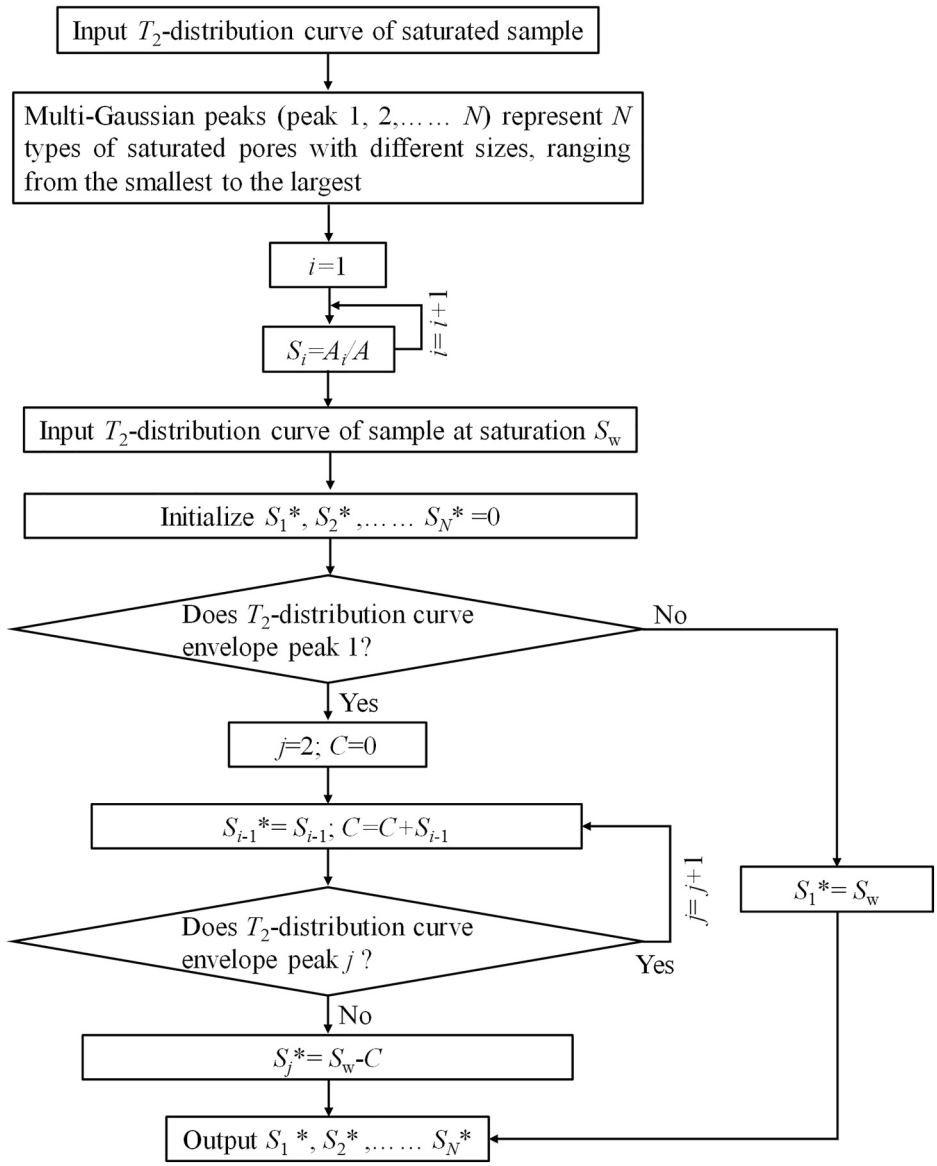


Figure 9. The flowchart for using T_2-distributions to determine the unsaturated water dynamics distribution of porous media (S_i is the contribution of pores, which is represented by peak i, to the saturation of saturated samples; A is the area enveloped by the saturated T_2-distribution curve, and A_i is the area enveloped by peak i; S_1*, S_2*,…… S_N* are the contribution of pores, which are represented by peak 1, 2,…… N, to the saturation of samples at saturation S_w).

real experimental data, and enables a more accurate interpretation of unsaturated NMR data. By understanding NMR responses in unsaturated media through simulation, we can effectively visualize and analyze pore structures and water distributions in both laboratory samples and field measurements (sNMR and bNMR). Field applications, especially in the vadose zone, often face challenges in achieving precise estimates of dynamic storage using NMR. To derive porosity from bNMR, as the water content and relaxation times can be very low, the commonly used multiexponential fit for fully saturated aquifers might lead to great uncertainty (e.g., discussions in Schmidt & Rempe, 2020). To mitigate these errors and address unsaturated NMR uncertainty, our analysis framework, using NMR measurement data from the unsaturated to saturated condition, can accurately evaluate water content and distribution in unsaturated soil and rock, monitor small changes in low water contents, and determine pore size and pore shape when NMR data from fully saturated conditions are also available.

Beyond soil and rock within the vadose zone, NMR has been used to characterize moisture adsorption, water vapor transport, and sap flow in natural materials such as plants, trees, wood, vegetables, fruits, and seeds. These

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systems are often composed of bio-cellulose materials containing water molecules with different mobilities (Cocusse et al., 2022; Zou et al., 2024). NMR has also been used to characterize concrete curing (e.g., Zheng et al., 2024) and determine the precipitation rate of cement hydration (e.g., Muller & Scrivener, 2017; Segura et al., 2023).

In many such studies, NMR results have indicated apparent over-saturation capacity in short relaxation times (e.g., Cocusse et al., 2022), as demonstrated earlier, highlighting the need for an accurate interpretation framework to investigate microporosity and delineate free water and bound water, and determine diffusion coefficients (e.g., Zou et al., 2023). Our method enhances the understanding of water transport in natural materials such as plants, trees, wood, vegetables, fruits, and seeds, and benefits the formulation, mixing, handling, and transportation of cellulose-based products. Such improved understanding of NMR results also supports efficient curing and hydration of cement-based building materials.

9. Conclusion

This paper aims to clarify how pore size and shape characteristics impact NMR relaxation behaviors in unsaturated porous media, and how to correctly evaluate pore water dynamic distribution from NMR responses of vadose zone rock and soil. Specifically, this paper develops a simulation framework to simulate unsaturated NMR responses of porous media and provides insights into interpreting complex architecture and water dynamic distribution of vadose zone soil and rock from NMR measurement data. The major conclusions are summarized below:

1. Because the small pores in the multi-pore system are saturated first, the enhanced NMR relaxation by surface relaxation in the unsaturated large pores induces large pores to display shorter relaxation times. After the large pores are saturated, surface relaxation is more difficult to enhance the NMR relaxation in large pores, which display longer relaxation times. Such shifts have been mistakenly attributed to increased water content in small pores within unsaturated media compared to their saturated status.
2. Pore shape affects the NMR relaxation behaviors in the unsaturated porous media due to surface relaxation. Even with a similar pore size distribution, the saturation of equal V_f/S_f of the small and large pores in the dual-triangular-shaped pore system is lower than that in the dual-spherical pore system. Because the equal V_f/S_f reflects the overlap of two relaxation time peaks representing water in the small and large pores, the evolution of T_2-distribution can reflect the pore shape inside rock or soil.
3. Pore size distribution affects the NMR relaxation behaviors in the unsaturated porous media due to surface relaxation. The saturation of the equal V_f/S_f decreases as the small pore size decreases, leading to the lower saturation of overlap of two relaxation time peaks. This variation can induce the misinterpretation of water dynamic distribution-small pores can contain more water than their maximum saturation capacity at lower saturation, implying the effect of pore size distribution on NMR relaxation behaviors should be considered in the interpretation of T_2-distribution in unsaturated pore-water dynamics.
4. A method is proposed to evaluate the pore-water dynamic distribution of unsaturated soil or rock by interpreting T_2-distribution. This pattern of pore water dynamic distribution ensures that the small pores do not exceed their maximum saturation capacity, avoiding common misinterpretations.

Appendix A: Simulation of T_2-Distribution of Unsaturated Porous Media

During the NMR experiment, when a radio frequency magnetic field (B_1) is applied, the Bloch-Torrey equation (Equation 7) can be written in the following matrix form:

$$\frac{\partial}{\partial t}\begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} + \begin{pmatrix} 1/T_{2B} & -\Delta\omega & \omega_x \\ \Delta\omega & 1/T_{2B} & -\omega_y \\ -\omega_x & \omega_y & 1/T_{1B} \end{pmatrix} - D\nabla^2 \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ m_0/T_{1B} \end{pmatrix} \tag{A1}$$

After the geometry model of unsaturated multi-pore media is established, the finite element method can be applied to solve the Bloch-Torrey equation. The physical parameters in Equation A1 refer to Oliveira (2021) and are exemplified by Table A1.

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Table A1
Physical Parameter in Equation A1

Physical parameter	Symbol	Value	Unit
Frequency offset	$\Delta\omega_1$	45	(Hz)
	$\Delta\omega_2$	0	
Coupling strength to the RF	ω_x (x-direction)	0	(Hz)
	ω_y (y-direction)	$\omega_y(t)$	
Bulk relaxation time	T_{1B} (Longitudinal direction)	3,000	(ms)
	T_{2B} (Transverse direction)	3,000	
Water bulk diffusion coefficient	D	2×10^{-9}	(m^2/s)
Equilibrium magnetization density	m_0	1×10^{12}	(A/m^4)

We set ω_y to a function $\omega_y(t)$ that varies with time t, simulating various pulse sequences to obtain different simulated outcomes from NMR experiments. To simulate T_2-distribution, $\omega_y(t)$ can be written as:

$$\omega_y = \begin{cases} \omega & 0 \le t \le \tau \text{ or } (2n-1)\tau_1 + (2n-1)\tau \le t \le (2n-1)\tau_1 + (2n+1)\tau \\ 0 & \tau < t < \tau + \tau_1 \text{ or } (2n-1)\tau_1 + (2n+1)\tau < t < (2n+1)\tau_1 + (2n+1)\tau \end{cases} \quad \text{(A2)}$$

where $n = 1, 2, 3, \ldots\ldots$ is the positive integer, ω is the coupling strength of pulse, τ is the duration of π/2-pulse, 2τ is the duration of π-pulse and τ_1 is the interval time of two pulses. The variation of $\omega_y(t)$ with time t is shown in Figure A1a. In practical measurements, the magnetic field is inhomogeneous, and the CPMG sequence uses a series of π-pulses to compensate for the effects of magnetic field inhomogeneities. However, in this study, because we assume an ideal external environment with a homogeneous magnetic field in the simulation, to reduce computation time and minimize the risk of non-convergence, a series of π-pulses are not used in the simulated pulse sequence. Equation A2 can be written as:

$$\omega_y = \begin{cases} \omega & 0 \le t \le \tau \\ 0 & \tau < t \end{cases} \quad \text{(A3)}$$

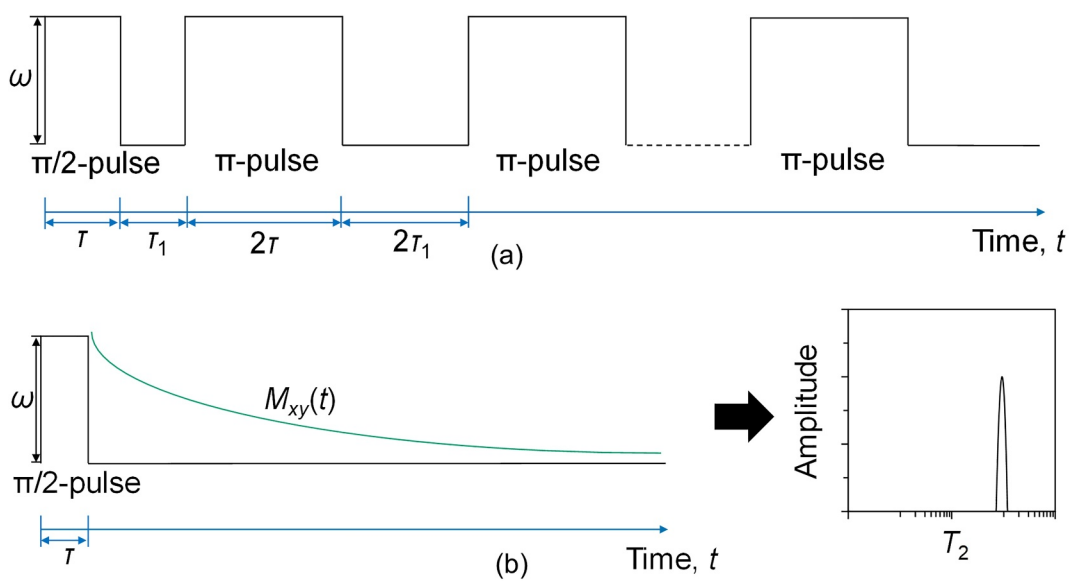


Figure A1. The variation of $\omega_y(t)$ with time t is utilized to simulate the CPMG pulse sequence: (a) the simulated CPMG pulse sequence; (b) the modified CPMG pulse sequence to simulate T_2-distribution.

The time-dependent variation $\omega_y(t)$ simulating the modified CPMG pulse sequence is shown in Figure A1b. In this study, we set up the coupling strength of pulse for ω = 0.45 MHz and the duration of π/2-pulse for τ = 3.4907 μs. The NMR observable is the time-dependent magnetization, obtained by integrating over the whole volume where the fluid is confined in the pore space:

$$M_{xy}(t) = \sqrt{\left[\int_V m_x(t)dV\right]^2 + \left[\int_V m_y(t)dV\right]^2} \tag{A4}$$

As shown in Figure A1b, the time-dependent variation of $M_{xy}(t)$ after removing the π/2-pulse qualifies the NMR signal decay of the transverse direction and can be inverted to the T_2-distribution.

Appendix B: Simulation of 2D T_1-T_2 Map of Unsaturated Porous Media

We use the function $\omega_y(t)$ that varies with time t to simulate the SR-CPMG sequence pulse and obtain simulated 2D T_1- T_2 distribution from NMR experiments. To simulate 2D T_1- T_2 distribution, $\omega_y(t)$ can be written as:

$$\omega_y = \begin{cases} \omega & 0 \le t \le \tau;\ \tau \le t - t_0 \le 2\tau;\ (2n-1)\tau_1 + 2n\tau \le t - t_0 \le (2n-1)\tau_1 + (2n+2)\tau \\ 0 & \tau < t < \tau + t_0;\ 2\tau < t - t_0 < 2\tau + \tau_1;\ (2n-1)\tau_1 + (2n+2)\tau \le t - t_0 \le (2n+1)\tau_1 + (2n+2)\tau \end{cases} \tag{B1}$$

where t_0 is the recovery time. The variation of $\omega_y(t)$ with time t is shown in Figure B1a. In this study, because the effect of magnetic field inhomogeneities can be neglected in the simulation, a series of π-pulses are not used in the simulated pulse sequence. Equation B1 can be written as:

$$\omega_y = \begin{cases} \omega & 0 \le t \le \tau \text{ or } \tau \le t - t_0 \le 2\tau \\ 0 & \tau < t < \tau + t_0 \text{ or } 2\tau < t - t_0 < 2\tau + \tau_1 \end{cases} \tag{B2}$$

The time-dependent variation of $\omega_y(t)$ simulating the modified SR-CPMG pulse sequence is shown in Figure B1b. The settings of ω and τ are the same as in the modified CPMG pulse sequence. As shown in Figure B1b, the time-dependent variation of $M_{xy}(t)$ after removing the second π/2-pulse qualifies that the NMR signal decays in the transverse direction under different t_0 and can be inverted to the 2D T_1-T_2 map.

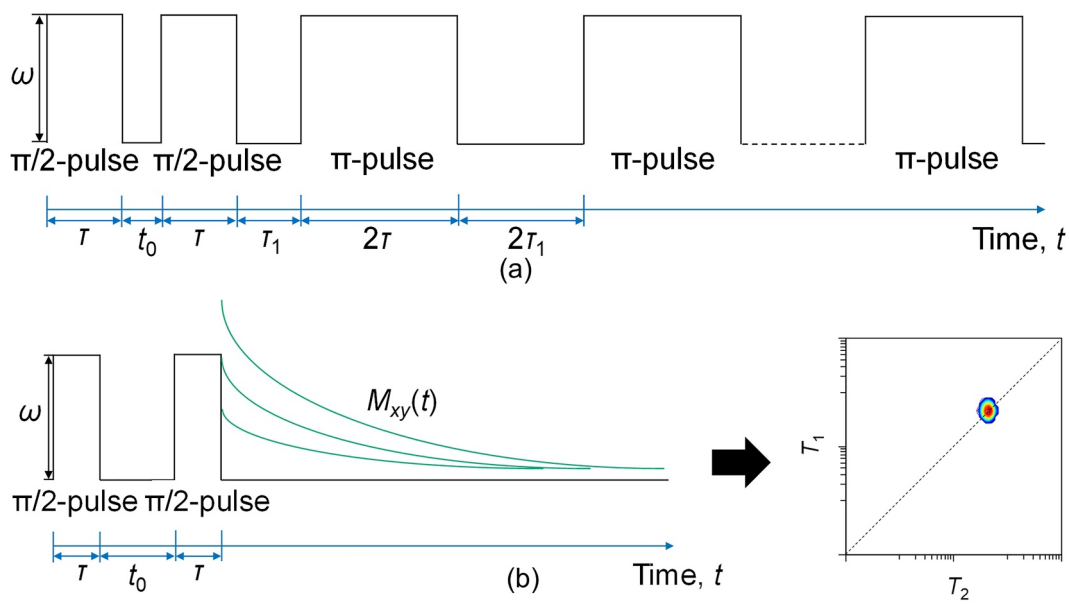


Figure B1. The variation of $\omega_y(t)$ with time t is utilized to simulate the SR-CPMG pulse sequence: (a) the simulated SR-CPMG pulse sequence; (b) the modified SR-CPMG pulse sequence to simulate 2D T_1-T_2 map.

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Data Availability Statement

All data of unsaturated NMR responses simulation in the single spherical pore and dual-systems in the article are available at Science Data Bank, via https://www.scidb.cn/en/s/RzIjqi, with CC-BY 4.0.

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文献 2/2

How Does Fluid Exchange Between Pores in Unsaturated Porous Media?

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How Does Fluid Exchange Between Pores in Unsaturated Porous Media?

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Key Points:

- We develop a framework that combines Nuclear magnetic resonance simulation and measurement to investigate pore coupling in unsaturated porous media
- Our simulation and measurement results reflect that the decrease in saturation weakens the degree of pore coupling
- Weakening pore coupling corresponds to more tortuous and eventually disconnected water-pathways due to reduced saturation

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The inter-pore fluid exchange, driven by molecular diffusion, is referred to as pore coupling. The pore coupling affects subsurface pore-scale physicochemical dynamics. Monitoring pore coupling is essential for understanding numerous hydrological and ecological processes. Nuclear magnetic resonance (NMR) has effectively tracked pore coupling in saturated porous media. However, monitoring unsaturated pore coupling with NMR is challenging due to the complex pore network architecture and dynamic water-pathway connectivity. Based on X-ray computed tomography imaging, we develop a framework to simulate the NMR T_2-distribution and T_2-store-T_2 map of samples with distinct-sized pore environments at different water saturations, validated against measurements. NMR results, combined with visualized pore networks and phase distributions, demonstrate that reduced saturation induces increasingly tortuous and eventually disconnected water-pathways, thus restricting and ultimately blocking pore coupling. Our work enhances NMR applications in unsaturated media, providing accurate insights into the complex architecture and hydrodynamics in the pore network.

Plain Language Summary Fluid is widely retained within the pore network in shallow vadose zone soils, weathered bedrock, and deep oil and gas formations. Monitoring inter-pore fluid diffusion and exchange (pore coupling) through the pore network is crucial for managing groundwater resources, preserving ecosystems, and mitigating environmental hazards. Nuclear magnetic resonance (NMR) is a powerful geophysical method to investigate pore-scale fluid dynamics. Although NMR can provide detailed insights into pore coupling within saturated soils and rocks, its application in unsaturated subsurface materials remains challenging, even at the laboratory scale. To address this challenge, we developed a framework to integrate measurements and simulations to track the pore coupling and its NMR response under the unsaturated state. First, using X-ray micro computed tomography, we visualize detailed pore network structures and phase distributions to simulate NMR results validated against measurements, investigating how pore coupling affects unsaturated NMR responses. The results also show that as saturation decreases, weakening pore coupling tracked by NMR detects more tortuous and eventually disconnected water-pathways, enabling the evaluation of connectivity in unsaturated pore networks by applying NMR to monitor pore coupling. Our framework enhances our understanding of how NMR can be applied in tracking hydrodynamics within a partially saturated subsurface environment.

1. Introduction

Extensive pore networks exist throughout subsurface porous media, from shallow vadose zone soils to deep aquifers and hydrocarbon reservoirs, governing underground fluid migration and storage. Within these pore networks, fluid molecules continuously diffuse and exchange between pores (e.g., Anand & Hirasaki, 2007; Grunewald & Knight, 2009; Toumelin et al., 2003), a process known as diffusional pore coupling. This inter-pore exchange influences pore-environment behaviors, enhancing pore-scale heat redistribution (Koch et al., 2021), promoting solute mixing (e.g., Baqer & Chen, 2022; Sadeghi et al., 2003; Jiménez-Martínez et al., 2020; Perez et al., 2023), and affecting pore-scale fluid-rock interactions by modifying local concentration gradients. Consequently, pore coupling alters key hydrodynamic and thermodynamic processes and regulates reaction kinetics within the subsurface. These effects influence natural processes such as freeze–thaw dynamics in permafrost aggregates (Rooney et al., 2022) and contaminant migration in the vadose zone (Chen et al., 2023), as well as engineered applications such as recovery and management of hydrocarbon reservoirs (Li et al., 2024).

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Despite its importance, the mechanisms controlling pore coupling remain poorly understood. Previous experimental studies have visualized diffusion-driven transport and its effects on pore-scale fluid kinetics in millifluidic or microfluidic devices with simplified pore systems using imaging techniques such as scanning electrochemical microscopy or high-speed camera (e.g., Jiménez-Martínez et al., 2015; Liu et al., 2016; Zhang et al., 2025; Zhou et al., 2022). Numerical simulations have also been applied to model pore-scale diffusion reactions based on idealized pore network models (e.g., Collin et al., 2025; Li et al., 2017). However, these idealized frameworks lack the complexity of natural pore network architectures and cannot capture the dynamic connectivity of water-pathways in partially saturated materials, where solid–liquid–vapor interactions strongly influence molecular diffusion. In particular, how the degree of pore coupling changes with decreasing saturation, and how this change can be observed and quantified, remains unresolved.

Low-field nuclear magnetic resonance (NMR) provides a powerful, non-invasive way to investigate fluid behavior within porous media due to its sensitivity to hydrogen in fluid molecules (Coates et al., 1999; Kirkland & Codd, 2018; Prammer, 2004). NMR measures magnetization and relaxation times (T_1 or T_2) of protons after an applied magnetic field removal to estimate water content, pore size distribution, and hydraulic conductivity (Behroozmand et al., 2015). In multimodal pore systems, pore coupling drives inter-pore exchange of magnetized protons, as shown in Figures 1a and 1b. This exchange averages NMR relaxation behavior (Soto Bravo & Zhang, 2023), causing T_2 peaks to shift or merge, which masks individual pore signatures and can make a multimodal system appear unimodal (Ramakrishnan et al., 1999). The 2D T_2-store-T_2 map captures this process through asymmetric off-diagonal peaks, which indicate magnetized proton exchange between pores and reflect both proton chemical shifts and inter-pore molecular exchange during mixing periods (Song, 2013; F. Zhang & Zhang, 2021). While NMR effectively tracks pore coupling in fully saturated materials (e.g., Anand & Hirasaki, 2007; Grunewald & Knight, 2009), variations in water-pathway geometry and connectivity complicate its application under unsaturated conditions. Accurately quantifying pore coupling and its NMR responses in unsaturated porous media is crucial to fully leverage NMR's potential for investigating pore-scale hydrodynamics and advancing understanding of subsurface physicochemical processes such as mineral dissolution and soil swelling.

To address this knowledge gap, we develop a framework combining NMR simulation and measurement with X-ray micro computed tomography (μCT) imaging to investigate pore coupling evolution in unsaturated multimodal-pore systems. Specifically, we: (a) integrate μCT imaging to visualize detailed pore network structures and phase distributions; (b) simulate NMR T_2-distributions and T_2-store-T_2 map at varying saturations and validate against laboratory measurements; (c) quantify how decreasing saturation weakens pore coupling by tracking increasingly tortuous and eventually disconnected water pathways. Our results substantially enhance NMR applications for characterizing unsaturated media and reveal how water-pathway connectivity governs pore coupling evolution, with implications for field-scale NMR investigations of subsurface hydrodynamics.

2. Experimental and Simulation Methods

2.1. NMR Theory and Relaxation Mechanisms

The spatiotemporal evolution of magnetization during NMR excitation and relaxation of protons in pore fluids is described by the Bloch-Torrey equation (Bloch, 1946; Torrey, 1956). The relaxation process manifests as transverse (T_2) or longitudinal (T_1) relaxation (Brownstein & Tarr, 1979), dominated by three mechanisms: bulk ($T_{1B,\ 2B}$), surface ($T_{1S,\ 2S}$), and diffusion relaxation (T_{2D}). These combined relaxation rate can be expressed as:

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1B,2B}} + \frac{1}{T_{1S,2S}} + \frac{1}{T_{2D}} \tag{1}$$

Bulk relaxation takes place spontaneously in the fluid and depends on fluid properties. Diffusion relaxation occurs in inhomogeneous magnetic fields and only affects T_2. Surface relaxation occurs in confined pore systems, and the relaxation rate of excited nuclei is enhanced by surface interactions at the fluid-solid interfaces, driven by the magnetic dipolar interactions between fluid spins and interfacial magnetic dipoles (Foley et al., 1996; Torrey, 1956). The active boundary condition is defined to reflect these molecular interactions at the solid-fluid boundary. In geologic media, the surface relaxation mechanism governs the relaxation process in the absence of magnetic field inhomogeneities. As a result, Equation 1 can be written as

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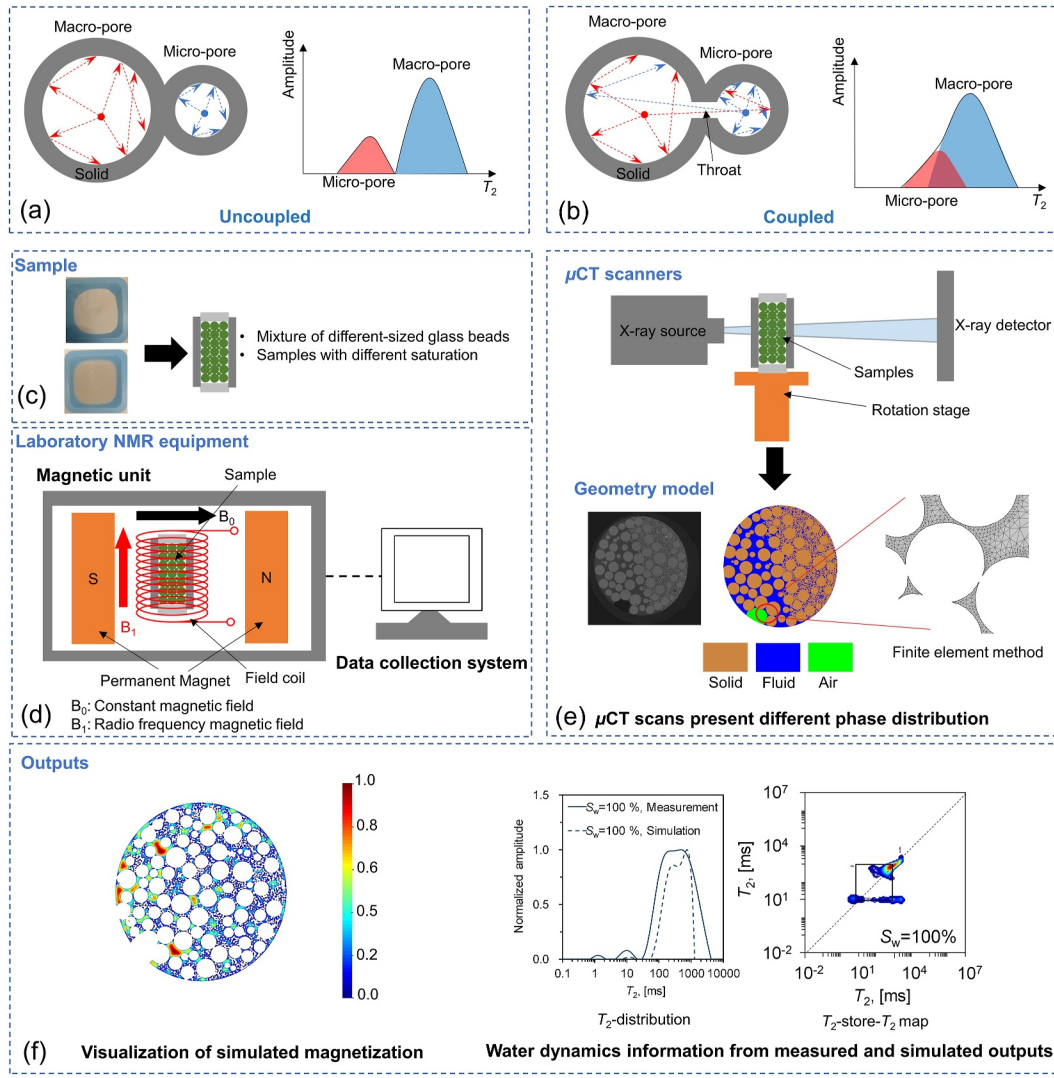


Figure 1. Scenarios of (a) uncoupled and (b) coupled pore system and their respective relaxation time distributions; schematic diagram of (c) Sample preparation and (d) Nuclear magnetic resonance (NMR) experiment, (e) numerical simulation of NMR based on μCT imaging, and (f) NMR outputs.

$$\frac{1}{T_{1,2}} \approx \frac{1}{T_{1S,2S}} = \rho_s \frac{S}{V} \tag{2}$$

Here, S represents the surface area of water-filled pores, ρ_s is the surface relaxivity, and V is fluid-filled pore volume. Equation 2 reflects a well-perceived correlation between NMR relaxation time and pore size. In the unsaturated system, as air is the non-contributing passive volume to NMR responses (Mohnke & Klitzsch, 2010), the passive boundary condition is defined to qualify the magnetization at fluid-air interfaces. For complete details on Bloch-Torrey equation, boundary conditions, and derivations, refer to Section 1. Theory of NMR simulation and measurement of Text S1 in Supporting Information S1.

2.2. NMR Measurement Protocol

Multimodal pore samples were prepared by mixing glass beads of varying sizes and packing them into thin-wall polypropylene tubes (10 mm inner diameter and 20 mm length). Each sample at a specific water saturation S_w was placed into a cylindrical holder and loaded into the 12 MHz NMR core analyzer from Niumag MacroMR (Figure 1c). Here, $S_w = V_w/V_{pore}$, where V_w is water volume and V_{pore} is total pore volume.

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Two types of NMR measurements were applied. First, we performed T_2 relaxation measurements on samples at different saturations using the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence. Subsequently, we conducted 2D T_2-store-T_2 measurements for these samples using Transverse Relaxation Exchange (TREx) pulse sequence. The TREx sequence consists of two CPMG trains separated by a storage period, during which the magnetization is stored along the z-axis (Montrazi et al., 2018). During this storage time, magnetized protons diffuse between pore environments, enabling inter-pore magnetization exchange. A 600 ms storage time was selected to ensure sufficient inter-pore proton exchange and thereby accurately and quantitatively capture the degree of pore coupling in multimodal systems. The collected NMR signals were inverted to obtain T_2-distributions and 2D T_2-store-T_2 maps of samples at different saturations. Complete details on pulse sequence parameters, and inversion methods are provided in Section 2. T_2-distribution simulation and measurement and 3. T_2-store-T_2 map simulation and measurement of Text S1 in Supporting Information S1, while details on sample preparation procedures are provided in Sample for NMR measurement of Text S2 in Supporting Information S1.

2.3. μCT Imaging and NMR Responses Simulation

The μCT imaging of samples was conducted using the HECTOR system (Masschaele et al., 2013) at the Ghent University Centre for X-ray Tomography (UGCT). The μCT setup is shown in Figure 1d. To achieve high-resolution imaging of phase distributions, smaller sample tubes (5 mm inner diameter and 20 mm length) were used compared to those for NMR measurements. Samples with different water saturations were prepared with 2.5% potassium iodide (KI) solution to enhance phase contrast. Sealed samples were scanned at 3 μm voxel resolution, and three-phase segmentation (solid, fluid, and air) was performed on reconstructed 2D grayscale cross-sections along the vertical height (Figure 1b). These segmented images form a 3D volume visualizing pore network structures and phase distribution in samples.

To reduce computational cost and uncertainty, pore network models were constructed directly from 2D phase-segmented images and used to simulate NMR responses via a commercial finite element software. Images of each sample were clustered by vertical position, which is identified by similar phase distribution and pore geometry. Each cluster containing an equal number of images represents an equal-length height range. One representative image from each cluster was selected for pore network model construction. The fluid-occupied space from selected images was meshed, incorporating boundary conditions at fluid–air and fluid–solid interfaces, to solve the Bloch–Torrey equation for time-dependent magnetization distribution. Temporal variations in the external magnetic field in the Bloch-Torrey equation were accounted for to accurately simulate both CPMG and TREx pulse sequences. Magnetization signals from ten representative simulations were superimposed and inverted to T_2-distributions and 2D T_2-store-T_2 maps. Sample information refers to Sample for μCT imaging of Text S2 in Supporting Information S1. Image processing procedures (Step 1 to 3), μCT imaging analysis (Step 4 and 5), and simulation parameters specifications (e.g., ρ_s calculated by Equation S11 and S12 in Supporting Information S1) (Step 4 and 5) are provided in Text S3 in Supporting Information S1.

3. Results and Discussions

We combined μCT-based NMR simulation with experimental validation to quantify how pore coupling evolves with saturation. Both T_2-distribution and T_2-store-T_2 map reveal progressive weakening of pore coupling as saturation decreases, corresponding to increasingly tortuous and eventually disconnected water-pathways. This framework provides quantitative metrics for tracking pore coupling in unsaturated multimodal-pore networks.

3.1. The Pore Coupling Effect on T_2-Distribution in Unsaturated Media

The T_2-distribution captures hydrogen relaxation signatures in the individual pore environment, with amplitudes reflecting their pore-water content. Inter-pore magnetization exchange distorts the T_2-distribution at different saturations, directly manifesting the pore coupling effect on NMR responses.

At full saturation ($S_w = 100\%$), both simulated and measured T_2-distribution show multiple peaks corresponding to large and small pores (Figure 2a). With decreasing saturation, the amplitude of the longest-relaxation time peak decreases more significantly than that of the shorter-relaxation components, reflecting the preferential moisture reduction within the largest pores (Figures 2c and 2d). Notably, at $S_w = 70\%$, the shorter-relaxation component amplitude exceeds that at $S_w = 100\%$ and 85%. This suggests that in the largest pores, unsaturated conditions enhance surface interactions, accelerating proton relaxation. As a result, some protons in large pores exhibit

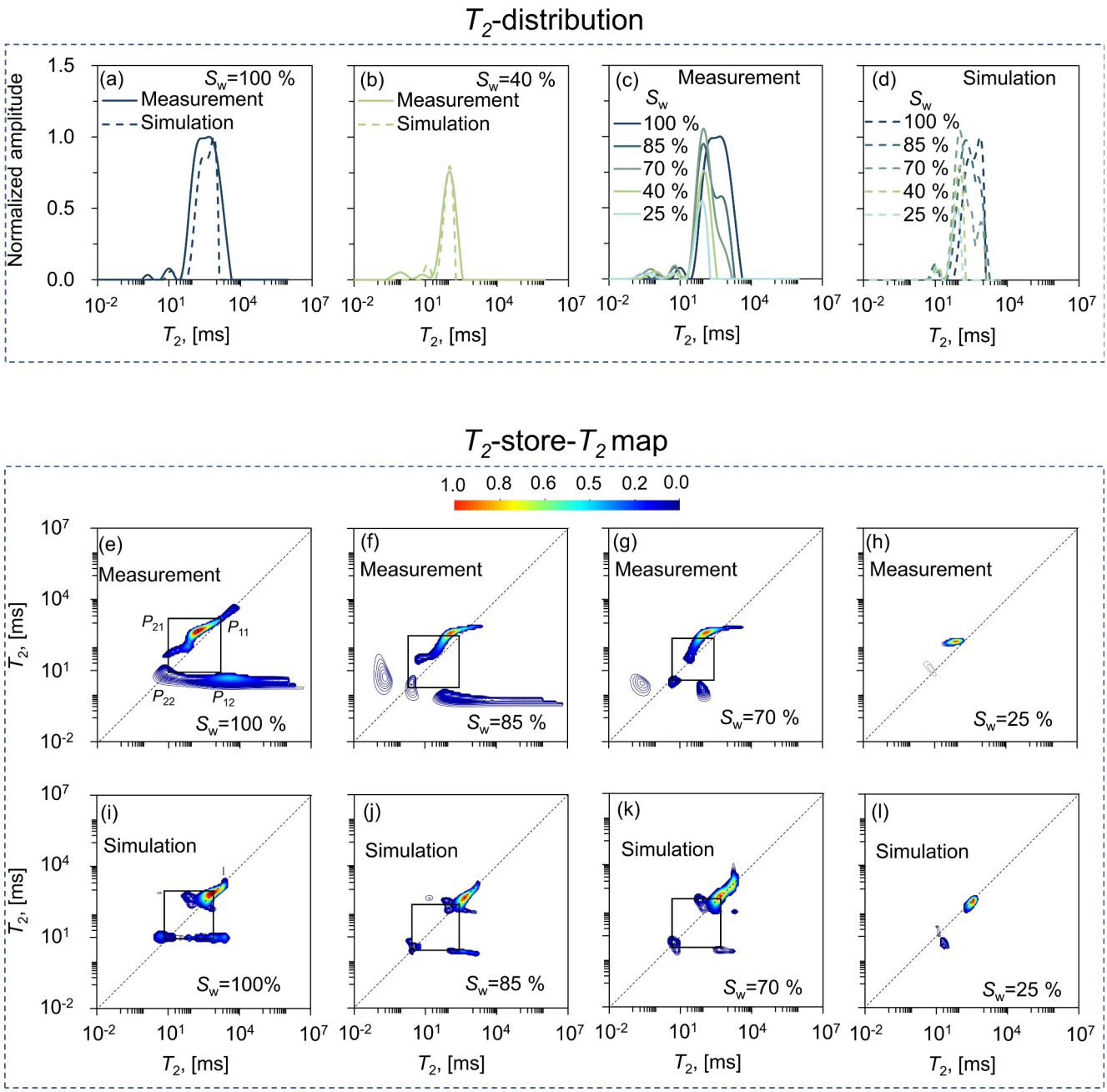


Figure 2. Measured and simulated T_2-distribution and T_2-store-T_2 maps of samples at different saturation states: Comparison of simulated and measured T_2-distributions at (a) S_w = 100%, (b) S_w = 40%; (c) Measured and (d) simulated T_2-distributions for samples at different saturations; Measured T_2-store-T_2 maps at (e) S_w = 100%, (f) S_w = 85%, (g) S_w = 70%, and (h) S_w = 25%; Simulated T_2-store-T_2 maps at (i) S_w = 100% (j) S_w = 85% (k) S_w = 70%, and (l) S_w = 25%.

relaxation behavior similar to those in smaller pores, contributing to the increased short-relaxation signal (J. Zhou & Zhang, 2025). As saturation decreases to 40%, the longest-relaxation time peak (largest pores) disappears entirely, leaving only shorter-relaxation components (Figure 2b), suggesting that water is retained only in small pores.

At the S_w = 100%, the two main relaxation time peaks nearly merged (Figures 2c and 2d). As saturation decreases, the peaks progressively separate, producing a clearer distinction between the long- and short-relaxation components at S_w = 85% and 70%. This comparison implies that under saturated conditions, the strong pore coupling facilitates inter-pore magnetization exchange, resulting in merged relaxation time peaks. As saturation

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decreases, pore coupling weakens, restricting inter-pore magnetization exchange and leading to clearer separation of the relaxation time peaks. While these observations directly reveal pore coupling evolution, the underlying mechanisms, such as the influence of pore channels' width, length, connectivity, and surface geochemistry (i.e. quantified by surface relaxivity (ρ_s) require further analysis. We will examine what factors weaken pore coupling in multi-pore systems as saturation decreases in the following discussion.

This weakened pore coupling effect on NMR relaxation can also arise from variations in pore network geometry and connectivity characteristics, including reduced pore channel width (e.g., Anand & Hirasaki, 2007), longer lengths (e.g., Mohnke & Klitzsch, 2010), and fewer connecting channels (Chi & Heidari, 2015). Mineral surface geochemistry, as quantified by surface relaxivity (ρ_s), can also regulate the inter-pore magnetization exchange. Higher ρ_s accelerates relaxation and thus weakens observable pore coupling effects (e.g., Grunewald & Knight, 2009). While our study focuses on geometric controls, ρ_s remains an important external factor that can influence NMR sensitivity to pore coupling. We further investigate the role of pore structure in governing coupling behavior as saturation decreases in the following discussion.

3.2. Quantitative Analysis of Unsaturated Pore Coupling From T_2-Store-T_2 Map

The T_2-store-T_2 map enables a qualitative assessment of inter-pore magnetization exchange, providing insight into the degree of pore coupling in unsaturated pore networks. In both measured (Figures 2e–2h)and simulated (Figures 2i–2l) T_2-store-T_2 maps, peak amplitudes are normalized by the maximum amplitude value in each map. The horizontal and vertical axes correspond to the T_2 relaxation times of the same proton population before and after exchange. Thus, in Figure 2e, diagonal peaks $P_{11,\,22}$ respectively represent magnetized protons remaining within large and small pore environments, while off-diagonal peaks $P_{12,\,21}$ correspond to protons diffusing from large to small pores and small to large pores, respectively (Schwartz et al., 2013). Monitoring these off-diagonal peaks enables quantitative estimation of pore coupling under saturated conditions (e.g., Fleury & Soualem, 2009; Song et al., 2014; Washburn & Callaghan, 2006).

Generally, an equal number of exchanged protons between coupled pores produces symmetric off-diagonal peaks about the diagonal ($T_2/T_2 = 1$). In this study, Figure 2 shows that only P_{12} appears in both simulated and measured T_2-store–T_2 maps, particularly at $S_w = 100\%$ and 85%. This asymmetry is attributed to the stronger surface relaxation in the small pore population, accelerating NMR relaxation of protons. Consequently, even if sufficient and balanced inter-pore exchange occurs, the magnetization carried by protons may decay before migrating from small to large pores, suppressing the NMR signal corresponding to P_{21}. This work utilizes the saturation-dependent evolution of P_{12} to monitor magnetized proton migration from large to small pores, thereby quantitatively tracking the progressive weakening of pore coupling with decreasing saturation.

Both measured (Figures 2e–2h) and simulated (Figures 2i–2l) T_2-store-T_2 maps exhibit that the amplitude of off-diagonal peak P_{12} decreases with reducing sample saturation. This evolution implies a decreasing proportion of magnetized protons migrating from the large to small pore population. Figure 3a shows the total amplitude of the off-diagonal peaks A_{od} relative to the total peaks' amplitude A_{tot} (A_{od}/A_{tot}) (Monteilhet et al., 2006), as a function of saturation. Both simulated and measured A_{od}/A_{tot} decline with decreasing saturation and approach zero at low saturation ($S_w = 25\%$), coinciding with the disappearance of the off-diagonal peak. This quantitative trend reflects the reduced proportion of inter-pore magnetized proton exchange and demonstrates that T_2-store–T_2 maps provide insight into quantitatively tracking the weakening pore coupling in multi-pore systems as saturation decreases.

3.3. Comparison Between Simulation and Experimental Results

This work establishes pore-network models based on 3D μCT imaging of real porous media to simulate NMR responses. First, we simulate T_2 distributions at different saturations to directly manifest the pore coupling effect on NMR responses. The simulation results are consistent with measurements, validating our framework. Compared with previous studies that only simulated T_2-store-T_2 maps for idealized saturated pore models (e.g., Fleury & Soualem, 2009), our framework simulates these maps to monitor the saturation-dependent pore coupling strength. While the simulated and measured T_2-distributions align well, some differences are observed in the peak shapes and relative positions of off-diagonal peaks in the T_2-store–T_2 maps. The observed discrepancies between the simulated and experimental T_2-store–T_2 maps mainly arise from two factors. For the simulation, NMR responses were obtained by superimposing results from 2D phase-segmented images at different

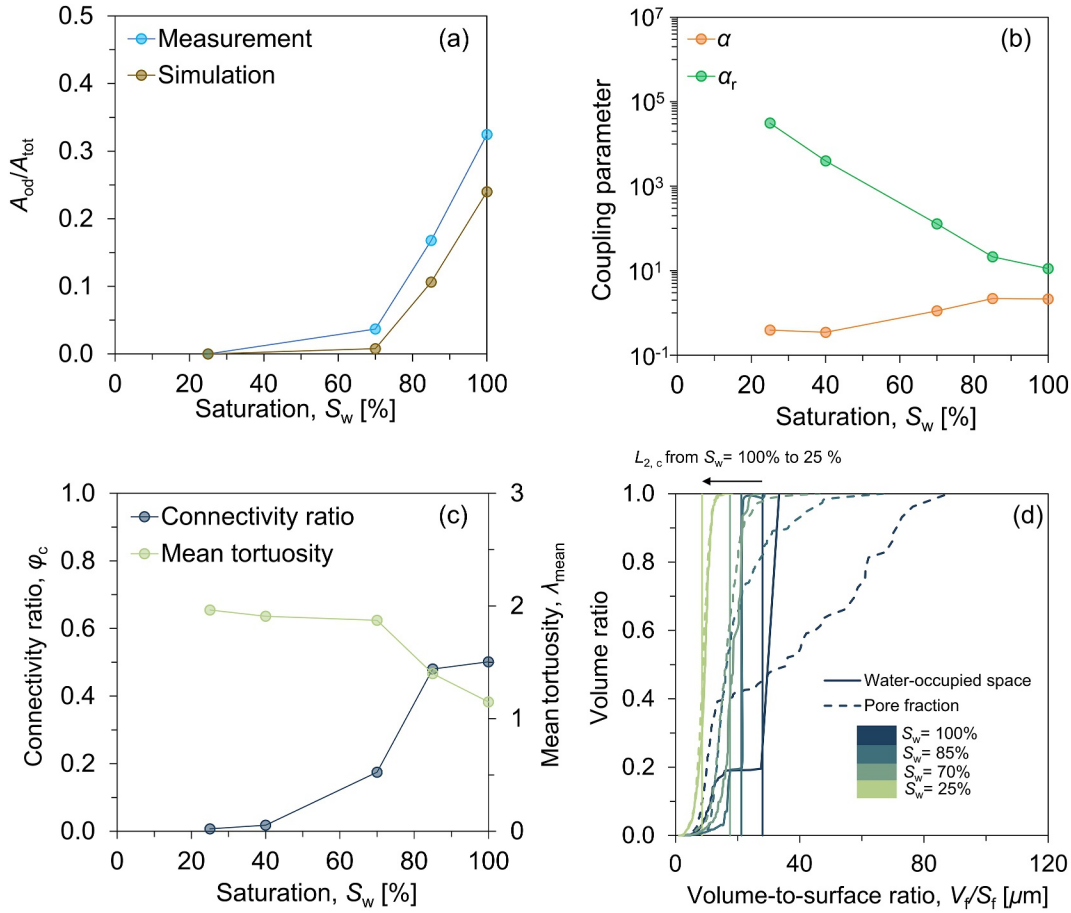


Figure 3. (a) The variation of A_{od}/A_{tot} during the desaturation process; The evolution of (b) α_r, (c) λ_{mean}, and φ_c during the desaturation process; (d) The cumulative distributions of V_f/S_f for connected pores and pore fractions at four saturations.

heights of the sample columns, which may potentially neglect vertical pore coupling. For the measurement, previous studies have presented that signal noise during data acquisition can cause deviations of off-diagonal peaks in inverted T_2-store–T_2 maps (via inverse Laplace transform) from the theoretical square positions formed with the diagonal peaks (Figures 2f and 2g) (e.g., Fleury & Soualem, 2009). This noise may also contribute to the trailing artifacts of the off-diagonal peaks observed in Figures 2e and 2f.

Despite these differences, the evolution of peak distribution and amplitude in simulated T_2-store-T_2 maps with varying saturations is broadly consistent with the experimental results. Specifically, the variation trend of A_{od}/A_{tot} with saturation derived from both simulation and measurement also matches well. Together, these results demonstrate that T_2-store–T_2 maps can be utilized to quantify pore coupling strength in pore networks with different structures and dynamic unsaturated pore-water distributions. Our simulation framework can predict NMR responses of unsaturated pore networks and provide insights into qualifying complex inter-pore water dynamics within subsurface porous media from NMR measurement data, thereby advancing our understanding of subsurface hydrological behaviors and processes.

3.4. Determinants of Pore Coupling Effect Under Unsaturated Conditions

In the following discussion, 3D μCT volumes of the samples are used to visualize the pore network structure and phase distribution. Based on the visualization, we elucidate factors responsible for the weakened pore coupling in multi-pore systems as saturation decreases, as revealed by the previous NMR simulation and measurement results. Multiple water-retained pores are interconnected by water-pathways, forming distinct water-occupied spaces within the unsaturated system. To demonstrate irregular water-pathways, the mean water-pathway tortuosity λ_{mean} is derived from skeletonized pathways in 3D μCT volumes. The effective diffusion coefficient is

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defined as $D_{\text{eff}} = D/\lambda_{\text{mean}}$. The connectivity for water movement is quantified by the connectivity ratio (φ_c), representing the probability that two randomly selected points belong to the same interconnected water-occupied region extracted from 3D μCT volumes. Details of λ_{mean} and φ_c are provided in the Step 5. Analysis of fluid distribution, fluid pathways tortuosity, and connectivity of Text S3 in Supporting Information S1. Surface relaxation occurs at the fluid-solid interface. The fluid volume (V_f^i) and surface area (S_f^i) of each connected space, and total values (V_f^{tot}, S_f^{tot}) of the entire system, are used to calculate the characteristic diffusion length $L_2^i = V_f^i/S_f^i$. L_2^i represents the average distance that surface-relaxing spins must diffuse throughout the space. Thereby quantifying the effective geometric size of the space. In the unsaturated system, Equation 2 can be expressed as $T_{2,c} = V_f^{\text{tot}}/\left(\rho_s S_f^{\text{tot}}\right)$ to define the characteristic relaxation time of system.

The coupling parameter $\alpha = \left(L_2/\sqrt{DT_{2,c}}\right)^2$, proposed by Anand and Hirasaki (2007), quantifies pore coupling effect on NMR relaxation in saturated systems with fully connected and regular pore networks. We propose the re-normalized coupling parameter of one connected space, expressed as $\alpha_r = \left(L_2/\varphi_c\sqrt{D_{\text{eff}}T_{2,c}}\right)^2$, to account for the effects of the surface geochemistry and irregular geometry of a partially connected water-pathway network in this space. The α_r of the entire unsaturated system can be expressed as

$$\alpha_r = \sum_i \frac{V_f^i\left[L_2^i/\left(\varphi_c\sqrt{D_{\text{eff}}T_{2,c}}\right)\right]^2}{\sum_i V_f^i} \tag{3}$$

to enable quantification of the pore coupling effect on NMR relaxation in the unsaturated pore network. Equation 3 indicates that magnetization decay in coupled-pore systems is governed by surface relaxation of spins and inter-pore exchange. When the diffusion length is much smaller than the diffusion distance within the characteristic relaxation time (i.e., $L_2^i \ll \varphi_c\sqrt{D_{\text{eff}}T_{2,c}}$), magnetized protons can diffuse between distinct pore environments before complete relaxation of their magnetization, leading to effective inter-pore magnetization exchange and thus strong pore coupling effect on NMR relaxation.

The α_r increases with decreasing saturation (Figure 3b). The system remains in an intermediate coupling regime ($1 < \alpha_r < 250$) for $S_w > 40\%$ (where $\alpha_r < 1$ represents total coupling in Anand & Hirasaki, 2007) and transitions to a decoupled state ($\alpha_r > 250$) at $S_w \leq 40\%$. This trend aligns with the weakening observable pore coupling qualified by the evolution of A_{od}/A_{tot} with saturation. Figure 3b also presents that while the α changes slightly, the α_r varies significantly with saturation. These evolutions imply that variations in saturation primarily alter the water-pathway tortuosity and inter-pore connectivity in the irregular and partially connected pore network, which may potentially influence the degree of pore coupling. Figure 3c further illustrates the evolution of λ_{mean} and φ_c with saturation. As saturation decreases from 100% to 85%, the λ_{mean} increases significantly while the φ_c changes slightly. This implies that although the unsaturated pores remain well-connected at $S_w = 85\%$, the reduced inter-pore channel water induces narrower and more tortuous water-pathways. When saturation further decreases from $S_w = 85\%$ to 70%, both λ_{mean} and φ_c change significantly, implying that some pathways become narrower and more tortuous, while others disconnect due to insufficient water. From $S_w = 70\%$ to 25%, φ_c decreases sharply while λ_{mean} remains nearly constant, suggesting that further moisture reduction no longer alters water-pathway geometry but directly blocks inter-pore connectivity.

The comparison of cumulative distributions of V_f/S_f in interconnected water-occupied spaces (dashed line) and individual unsaturated pore fractions (dashed line) at four saturations (Figure 3d) also illustrates how varying saturation influences the inter-pore connectivity. $L_{2,c} = V_f^{\text{tot}}/S_f^{\text{tot}}$ reflects the characteristic length for surface-relaxing spins to diffuse throughout the system under the total coupling state (Fleury & Soualem, 2009). At high saturation, the two curves differ significantly, with pore fractions exhibiting a much larger maximum V_f/S_f. This indicates that water-occupied spaces are formed by the connection of numerous unsaturated pores, with the effective geometry size of spaces represented by the average characteristic size of pores forming them. Especially, at $S_w = 100\%$ and 85%, the water-occupied space curve displays a sharp rise near $V_f/S_f = L_{2,c}$, implying that most unsaturated pores are well-connected to form a main water-occupied space. As saturation decreases, two curves gradually evolve toward convergence. These pores no longer form a single interconnected space but segregate into isolated smaller pore clusters with limited connectivity, due to partial disconnection of water-

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pathways. At low saturation, most pores become completely isolated, forming numerous independent water-occupied spaces.

The aforementioned evolution illustrates how saturation-induced variations in water-pathway geometry and inter-pore connectivity govern the pore coupling strength. As saturation decreases, reduced water content in inter-pore channels induces narrower and more tortuous pathways. This variation results in longer fluid diffusion paths, thereby restricting the inter-pore molecular exchange. At low saturation, insufficient water fails to maintain inter-pore connectivity. The interconnected water-occupied space progressively segregates into several smaller pore clusters with limited connectivity. Eventually, these clusters fragments into isolated and individual pore environments as most water-pathways become completely disconnected. Consequently, the inter-pore fluid diffusion is blocked, eliminating the pore coupling.

4. Geophysical Implications and Broader Applications

Our results demonstrate that decreasing water saturation leads to progressively weaker pore coupling due to increased tortuosity and disconnection of water pathways. This process is directly reflected in the merging or broadening of T_2 peaks and the loss of off-diagonal features in T_2–store–T_2 maps. This relationship has broad implications for interpreting NMR data in hydrological models that can be used for groundwater management and environmental remediation.

By quantifying coupling strength under variable saturation, NMR can inform parameterizations in dual-porosity or mobile-immobile domain models, especially in predicting the retention, release, or accessibility of water and solutes across pore scales. Furthermore, common NMR-derived permeability estimation models such as the Schlumberger Doll Research (Kenyon et al., 1988) and Timur-Coates (Coates et al., 1999; Timur, 1969) models are calibrated for saturated conditions and may significantly misestimate hydraulic conductivity in unsaturated vadose zones. While several previous studies have applied NMR-derived parameters to estimate subsurface hydraulic conductivity in unsaturated media (e.g., Legchenko et al., 2004; Legchenko & Vouillamoz, 2011; Vouillamoz et al., 2003), these approaches generally rely on averaged relaxation properties and do not explicitly account for pore coupling, which reflects saturation-dependent inter-pore connectivity. As a result, permeability estimates may be inaccurate under partially saturated conditions. The degree of pore coupling, quantified through A_{od}/A_{tot} and α_r, provides correction factors for estimating hydraulic conductivity in partially saturated conditions. Field NMR instruments, including surface NMR and borehole NMR, detect T_2-distribution profiles along depth in the natural subsurface (e.g., Guérin, 2005; Jiang et al., 2020; Kirkland & Codd, 2018). Our findings clarify that vertical variations in T_2 spectra might reflect pore coupling evolution due to saturation changes. This distinction enhances field data interpretation, particularly in heterogeneous formations where saturation gradients are strong.

Beyond NMR, other geophysical methods such as spectral induced polarization (SIP) are sensitive to pore-scale connectivity and diffusion-controlled charge transport (Tsukanov et al., 2021), the same processes quantified through our pore coupling framework. Integrating NMR measurements with SIP and other geophysical methods enables more robust characterization of subsurface fluid dynamics and reactive transport, improving predictive models of water–rock interactions in complex geologic environments.

5. Conclusion

This study aims to clarify how NMR responds to the pore coupling in unsaturated porous media, and how variation in saturation governs the pore coupling strength. Specifically, this paper develops a framework combining simulation and measurement to investigate pore coupling and its NMR responses in unsaturated porous media and provides insights into investigating hydrodynamics within complex subsurface geologic environments through NMR observation.

Realistic pore-network models constructed from 3D μCT imaging are used to simulate the evolution of T_2-distribution and T_2-store-T_2 map from saturated to unsaturated states, validated against NMR measurement results. The evolution of T_2-distribution reveals that the decreased saturation weakens the pore coupling effect on NMR relaxation. This weakened pore coupling strength can be quantitatively tracked by T_2-store-T_2 maps.

Based on μCT imaging, we introduce the mean water-pathway tortuosity λ_{mean} and connectivity ratio φ_c into the re-normalized coupling parameter α_r. The α_r quantitatively evaluated the attenuation of the degree of pore coupling effect on NMR relaxation with saturation decreasing. The evolution of λ_{mean} and φ_c, combined with the

comparison of cumulative distribution curves of V_f/S_f for interconnected water-occupied space and individual pore fractions at different saturations, clarify the factors responsible for weakened pore coupling: from fully saturated state to high saturation condition, water-pathways become increasingly tortuous, as indicated by the rise in λ_{mean}, which restrict the inter-pore fluid exchange; as saturation further decreases, these tortuous pathways progressively disconnect, and the interconnected water-occupied space progressively segregates into numerous independent unsaturated pores, leading to a sharp decline in φ_c and ultimately blocking pore coupling. This mechanism suggests that NMR tools can be used to monitor the degree of pore coupling, thereby enabling the evaluation of pore-scale connectivity and diffusion-controlled transport under unsaturated conditions.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All data of CT images, NMR responses simulation, and NMR measurement in the article are available at Science Data Bank, via J. Zhou (2025), with CC-BY 4.0.

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